

Full Guide

Every menu, field, formula, model activity and analysis path

What this workbench does

It turns a mission into a connected early engineering model: stakeholder needs and objectives lead to requirements; functions and components describe the solution; parameters, verification and simulations provide evidence; configurable alternatives can be compared and approved. It supports preliminary programme and architecture work, not certified detailed-design analysis.

Two synchronized ways to work

Architect view One guided question at a time, with answer guidance, examples, progress and readiness.

Modeler view Dashboard plus direct tables, graphs, matrices, formulas, variability, analysis and delivery.

Both views update the same canonical project; switching never creates a second model.

Choose a scope

Architecture definition	Traceable product and industrial-system architecture
Architecture and simulation	Architecture plus KPI evidence for the current model
Trade-off	Configurable 100% alternatives, comparable runs and a decision

Complete reference for the current Architect and Modeler perspectives.

MBSE / MBPLE Workbench - application v1.8.0 - schema 14

Guide refreshed 25 September 2026. The introduction occupies this single page.

Continue with the existing guide, then use the appended answer handbook for every Architect question group.

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How to use this guide

This is the complete operating manual for the MBSE / MBPLE Programme Workbench. Read Part I once, then use the workspace chapters as a field reference. Part X contains a coffee-machine build-from-scratch walkthrough and complete recipes for all three scopes.

Authority of this guide

The current webpage is the master. This guide describes the behavior that is present in workflow v1.5.3. A field marked documentation does not secretly participate in a calculation. A placeholder is only a hint and is never stored unless you type it.

Effect labels used throughout

Label	Meaning	Typical examples
RUN	Directly changes a calculation, requirement result, derivation, simulation, comparison or baseline.	Parameter value, unit, formula, process duration, feature selection, KPI weight.
MODEL	Changes canonical semantics, traceability, scope, eligibility or completeness.	System of interest, working use case, relationship type, realization link.
GOV	Changes maturity, validation, provenance or approval evidence.	Status, source, owner, decision rationale.
DOC	Stored and exported for human understanding but not used numerically.	Description, relationship name, axis description.
VIEW	Changes only what you see or the order in which it is shown.	Search, filters, layout, audience view.
READ	Calculated or historical information. You inspect it but do not type into it.	Revision, run result, derivation audit, normalized score.
HINT	Placeholder example inside an empty control. It has no effect until replaced by typed text.	@mass <= 1000, kg, Optional flow name.

The complete evidence chain

Question	Model answer	Why the connection matters
Why does the project exist?	Mission -> stakeholders -> needs and objectives	Defines the intent from which scope and requirements are justified.
What behavior is in scope?	System of interest + involvedIn + selected use cases	Limits architecture and trade-study evidence to the intended investigation.
What must be true?	Need/objective -> derives -> system requirement	Makes mandatory feasibility traceable to stakeholder intent.
What does the product do?	Use case -> hasFunction -> product function	Defines product behavior before structure.
What realizes product behavior?	Product function -> realizedBy -> product component	Connects functions to the product technical architecture.
How is it produced?	Use case -> process function -> industrial system component	Connects manufacturing behavior to the industrial technical architecture.
What resources are required?	Industrial component -> requiresResource -> resource	Feeds process cost and resource demand using the realizing component.
What numeric evidence exists?	Parameters, requirement formulas, KPI definitions	Provides the values and rules used by simulation and feasibility.
What alternative is evaluated?	150% model -> feature selection -> 100% realization	Creates variant-specific architecture evidence without mutating the family model.
What decision is supported?	Current simulations -> comparison -> approved decision	Preserves feasibility, scoring, evidence and rationale.

Resource allocation rule

For every scope, store requiresResource from the industrial-system component to the Resource. Do not allocate a resource directly to the process function. The process function is realizedBy the industrial component; the component requires the resource.

Part I - Global page controls and project lifecycle

The header and left navigation remain available while you move between workspaces. They control the active project and the context in which the model is displayed.

Perspective selection

The opening dialog selects an authoring perspective. Both perspectives operate on the same active project; switching perspective never creates a second model.

Field or control	What to enter or select	Allowed values / rule	Actual effect
Architect view	Click to build or review the project through one question at a time.	Available for every project and all three scopes.	MODEL. Confirmed answers create or link canonical elements, relationships, parameters, formulas and analysis evidence.
Modeler view	Click to open the existing dashboard and detailed workspaces.	Available for every project.	VIEW. Opens direct model editing; later edits can mark guided answers for review.

Architect view question interface

Action	What happens	Use it when	Impact
Continue	Validates the current answer, writes its model effect and opens the next applicable question.	After entering or selecting a complete answer.	MODEL, RUN or GOV according to the question.
Back	Returns to the previous applicable question.	Reviewing or correcting recent input.	VIEW until an answer is changed and confirmed.
Skip	Records a visible gap and opens the next question. There is no timer or automatic selection.	Only when information is genuinely unavailable.	GOV. Required skipped content prevents Ready status.
Progress section	Opens the first incomplete question in that section and shows answered/skipped counts.	Navigating a long session or resuming work.	VIEW.
Open Modeler view	Switches to the detailed dashboard and editors using the same project.	Inspecting matrices, formulas, graphs or advanced evidence.	VIEW; no project content is duplicated.

Architect view generation and review rules

- Stakeholders are defined before the system of interest is selected.
- Quantitative requirements follow use-case definition in every scope. The operator, target, unit, parameter source and engineering value are confirmed before requirement review.
- Need/objective -> requirement traceability and requirement -> satisfying function, component or parameter evidence are created explicitly.
- Every requirement needs at least one satisfaction path; not every function or component needs a requirement link.
- For a trade-off study, the decision question and common/alternative solution characteristics follow requirement definition, before product architecture.
- Consumed and produced product flows are selected from product components already defined in the product structure stage.

- Changing generated content shows an impact review. Direct Modeler edits mark prior guided answers for review instead of silently accepting stale evidence.

Architect view final recaps

Recap	Shown for	Minimum content
Study overview	All scopes	Purpose, selected scope and scoped use cases.
Stakeholder traceability	All scopes	Stakeholders, needs/objectives and quantitative requirements.
Architecture	All scopes	Product/industrial functions and technical components.
Requirements	All scopes	Satisfied, failed, pending and error counts.
Simulation	Simulation and trade-off	Current run, KPI count, warnings and assumptions.
Trade-off	Trade-off only	Alternatives, KPIs, comparison notes and proposed/approved baseline.

Left navigation

Action	What happens	Use it when	Impact
Dashboard	Opens the selected-scope workflow and live completion state.	Starting work or deciding what to do next.	VIEW plus MODEL guidance; does not create data by itself.
Problem Space / Solution Space steps	Opens the first activity for that workflow step and records a return target.	Following the recommended storyline.	VIEW only; evidence in the target workspace determines completion.
Model	Opens all MBSE elements, relationships, formulas, matrices, sequences and validation.	Creating or correcting the canonical 150% model.	MODEL, RUN and GOV depending on the edit.
Variability	Opens features, variation points, configurations and derived 100% models.	Building product-line variability or variants.	MODEL and RUN.
Parameters and KPIs	Opens engineering inputs and reusable KPI definitions.	Preparing calculations and simulations.	RUN and GOV.
Simulation	Runs selected KPI definitions or direct algorithms and retains immutable results.	Producing comparable evidence.	RUN.
Architecture Trade Study	Frames, compares and decides among current configuration-derived alternatives.	Selecting a baseline.	RUN and GOV.
Export	Saves, loads, snapshots and exports evidence.	Recovery, review and delivery.	GOV and persistence.

Header controls

Field or control	What to enter or select	Allowed values / rule	Actual effect
Active project	Read the project currently being edited.	Read-only name.	READ. Every workspace acts on this project.
Project selector	Choose another locally stored project.	Any project in this browser storage.	MODEL context changes; no project data is merged.
New project	Click, then type a project name into the browser prompt.	Non-empty text. The initial hint is New engineering study.	MODEL. Creates a clean project and makes it active.
Duplicate	Click once while the intended source project is active.	No input field.	MODEL. Creates an independent copy with new identity.
Delete	Click and confirm deletion of the active project.	Confirmation required.	MODEL. Removes the local project; use a saved JSON if recovery may be needed.
Save Project	Click to download the complete active project JSON.	No options in the header action.	GOV. Portable reloadable project evidence.
Load Project	Click to open Export > Save / Load Project.	File and load behavior are chosen there.	VIEW navigation only until Validate and load is clicked.
Architecture context	Select the 150% model or an available configuration / architecture.	150% Product-Line Model or non-archived architecture.	VIEW and analysis context. It filters model evidence but does not rewrite the source.
Local autosave	Read Saved, Saving or Save error.	Read-only state.	READ. Browser-local continuity, not enterprise version control.
Revision	Read the current model revision.	Positive internal counter.	READ. Used to identify stale derivations, simulations and comparisons.
Reset sample	Click and confirm.	Active sample project only.	MODEL. Recreates the sample content; unsaved changes are lost.
Reset all	Click and confirm.	All locally stored projects.	MODEL. Clears local application data and recreates defaults.
Return to workflow	Click after entering a workspace from a dashboard activity.	Shown only when workflow context exists.	VIEW. Returns to the selected dashboard step.

MBSE / MBPLE
Programme workbench

ACTIVE PROJECT
Configurable Coffee Machine Product Line Configurable Coffee Machine Product L + New project Duplicate

Save Project Load Project 150% Product-Line Model

Local autosave: Saved Revision 1 Reset sample Reset all

Snapshots and Delivery Stage C

Create basic local recovery points and export validated, scoped engineering evidence.

Preliminary engineering estimate — not a verified detailed-design result.

Save / Load Project Snapshots Selective Export PDF Report Demo Checklist

Save Project
Download the complete active project as a portable, validated JSON package.

Configurable Coffee Machine Product Line
Schema 9 · revision 1 · 41 model elements

The project file includes model, variability, configurations, derivations, simulation evidence, Trade Studies and decisions. Application-level snapshots and interface preferences remain local and are not embedded.

Download project file

Load Project
Validate a saved project file before loading it. Invalid input changes nothing and remains available for raw download.

SAVED PROJECT FILE
 No file chosen

LOAD BEHAVIOR

☒ **Load as new project**
Keep the active project and resolve only a top-level project-ID collision.

☐ **Replace active project**
Preserve the active project ID and create a safety snapshot first.

Validate and load

Global controls select the active project, save or load the complete model, choose a 150% or configured context, and show autosave plus revision state.

Safe project-start procedure

1 Create or duplicate

Click New project for a blank model. Type Coffee Machine Tutorial and press OK. Use Duplicate only when you want to preserve the current sample as a starting template.

Why: A project identity separates later edits, simulations and exports.

Done when: The new name appears under Active project.

2 Choose the overall scope

Open Dashboard and select exactly one overall-scope radio card.

Why: The selected scope changes the storyline while preserving all project data.

Done when: The dashboard badge shows the intended scope.

3 Create a portable checkpoint

Click Save Project. Store the downloaded JSON with a date or milestone in its filename.

Why: Browser autosave is convenient; the downloaded JSON is the deliberate transfer and recovery artifact.

Done when: A JSON file is downloaded.

4 Work on the canonical context

Keep Architecture context at 150% Product-Line Model while editing reusable source content.

Why: Configured views are projections; the 150% model is the canonical family source.

Done when: The header shows 150% Product-Line Model.

Placeholder rule

Faint example text inside a field is only a suggestion. It is neither saved nor evaluated. Click the field and type the intended value.

Part II - Dashboard and the three master storylines

The dashboard is the master route map. It derives every status from real project evidence. You do not manually mark a step complete.

Dashboard controls and status

Field or control	What to enter or select	Allowed values / rule	Actual effect
Overall scope	Select Architecture building, Architecture building + simulation, or Trade study.	Exactly one radio option.	MODEL storyline. Existing data is preserved; completion is recalculated.
Workflow step card	Click the step you want to inspect.	Any card in Problem Space or Solution Space.	VIEW. Opens live status, message and activity buttons.
Activity button	Click a labeled activity such as Product functions or Parameters and KPIs.	Buttons are determined by scope and step.	VIEW. Opens the correct workspace, tab, view and type filter.
Active Trade Study	For the trade-study definition step, select the intended study.	Any stored Trade Study.	MODEL context. The active study drives scope, alternatives and completion.
Status badge	Read Not started, In progress, Complete or Blocked.	Read-only derived state.	READ. Complete means required evidence exists, not that a user checked a box.
Recommended next	Follow the first non-complete step unless intentionally diagnosing later content.	Read-only guidance.	VIEW guidance.

Scope 1 - Architecture building

Phase	Step	Required evidence
Problem Space	Define mission, system and intent	Mission, mission stakeholder, exactly one system of interest, at least one need and objective.
Problem Space	Define use-case and requirement scope	A selected eligible use case and at least one requirement derived from scoped need or objective.
Solution Space	Build the product functional & technical architecture	Product functions connected from each selected use case; every product function realized by a product component.
Solution Space	Build the industrial system functional & technical architecture	Industrial system functions connected from each selected use case; every function realized by an industrial-system component.

The screenshot displays the MBSE / MBPLE Workbench interface. On the left is a dark sidebar with a 'Dashboard' button and sections for 'PROBLEM SPACE' (2/2 steps), 'SOLUTION SPACE' (2/2 steps), and 'WORKSPACES' (Model, Variability, Parameters and KPIs, Simulation, Architecture Trade Study, Export). The main area shows the 'ACTIVE PROJECT' 'Configurable Coffee Machine Product Line' with a dropdown menu set to 'Configurable Coffee Machine Product L'. Below this are buttons for 'Save Project', 'Load Project', and a '150% Product-Line Model' dropdown. A status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'. The 'PROJECT WORKFLOW' section is titled 'Configurable Coffee Machine Product Line' and includes a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' It instructs users to 'Choose the project outcome, then complete the model activities shown in Problem Space and Solution Space.' Under 'Overall scope', three options are listed: 'Architecture building' (selected), 'Architecture building + simulation', and 'Trade study'. A status bar shows 'Status: Not started, In progress, Complete, Blocked'. The 'Problem Space' section (2/2 complete) shows two steps: '1 Define mission, system and intent' and '2 Define use-case and requirement scope'.

Architecture building uses the product architecture step followed by the industrial-system architecture step.

Scope 2 - Architecture building + simulation

Phase	Step	Required evidence
Problem Space	Define mission, system and intent	Same traceable intent as Architecture building.
Problem Space	Define use-case and requirement scope	Same selected semantic scope and derived requirements.
Problem Space	Define evaluation KPIs	At least one KPI; every KPI has valid non-negative weight and at least one objective.
Solution Space	Build product functional & technical architecture	Product functions and realizing product components.
Solution Space	Build industrial system functional & technical architecture	Process functions and realizing industrial components.
Solution Space	Complete architecture parameters	At least one parameter and a value for every numeric parameter required in the active model.
Solution Space	Simulate architecture	At least one current saved simulation run.

The screenshot displays the MBSE / MBPLE Programme workbench interface. The left sidebar contains a navigation menu with sections for 'PROBLEM SPACE' (3/3 steps), 'SOLUTION SPACE' (4/4 steps), and 'WORKSPACES' (Model, Variability, Parameters and KPIs, Simulation, Architecture Trade Study, Export). The main workspace shows the 'ACTIVE PROJECT' 'Configurable Coffee Machine Product Line' with a dropdown menu set to 'Configurable Coffee Machine Product L'. Below this are buttons for 'Save Project', 'Load Project', and a dropdown for '150% Product-Line Model'. A status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'. The 'PROJECT WORKFLOW' section is titled 'Configurable Coffee Machine Product Line' and includes a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' Below this, the 'Overall scope' section shows three options: 'Architecture building' (Not started), 'Architecture building + simulation' (Selected, In progress), and 'Trade study' (Not started). The 'Problem Space' section shows three steps: '1 Define mission, system and intent', '2 Define use-case and requirement scope', and '3 Define evaluation KPIs' (3/3 complete).

Architecture building + simulation adds KPI definition, parameter completion and a current saved run.

Scope 3 - Trade study

Phase	Step	Required evidence
Problem Space	Define mission, system and intent	Traceable intent and exactly one system of interest.
Problem Space	Define use-case and requirement scope	Eligible project working scope and derived requirements.
Problem Space	Define trade study, variability axes and evaluation KPIs	Study name/question, scoped use cases/needs/objectives, mandatory requirements, one root, selected axis and valid objective-linked KPIs.
Solution Space	Build the 150% architecture	Product and industrial-system functions plus technical realization links in one reusable family model.
Solution Space	Define feature model and constraints	Exactly one root and selectable non-root feature content.
Solution Space	Map variability to architecture	At least one valid variation point.
Solution Space	Create valid configurations	At least two active valid configurations.
Solution Space	Derive 100% architectures	At least two current derivations from the current revision.
Solution Space	Simulate architectures	At least two current complete simulations covering all selected KPIs.
Solution Space	Compare and decide	A current immutable comparison and an approved decision linked to the study.

The screenshot displays the MBSE / MBPLE Programme workbench interface. On the left is a dark sidebar with a 'Dashboard' button and two main sections: 'PROBLEM SPACE' (3/3) and 'SOLUTION SPACE' (7/7). The 'PROBLEM SPACE' section lists three tasks: 'Define mission, system and intent', 'Define use-case and requirement scope', and 'Define trade study, variability axes and evaluation KPIs'. The 'SOLUTION SPACE' section lists seven tasks: 'Build the 150% architecture', 'Define the feature model and constraints', 'Map variability to architecture', 'Create valid configurations', 'Derive 100% architectures', 'Simulate architectures', and 'Compare and decide'. Below these are 'WORKSPACES' for 'Model', 'Variability', and 'Parameters and KPIs'.

The main area shows the 'ACTIVE PROJECT' 'Configurable Coffee Machine Product Line'. It includes buttons for 'Save Project', 'Load Project', and a dropdown for '150% Product-Line Model'. A status bar indicates 'Local autosave: Saved', 'Revision 1', and buttons for 'Reset sample' and 'Reset all'. The 'PROJECT WORKFLOW' section is titled 'Configurable Coffee Machine Product Line' with a warning: 'Preliminary engineering estimate — not a verified detailed-design result.' Below this, it instructs to 'Choose the project outcome, then complete the model activities shown in Problem Space and Solution Space.'

The 'Overall scope' section offers three options: 'Architecture building' (Define the problem and build the functional, product and industrial architecture), 'Architecture building + simulation' (Build one architecture, complete its parameters and simulate it), and 'Trade study' (Build configurable alternatives, simulate architectures, compare and decide). The 'Trade study' option is selected. A status bar shows 'Not started', 'In progress', 'Complete', and 'Blocked'.

The 'Problem Space' section is marked '3/3 complete' and contains three steps: '1 Define mission, system and intent', '2 Define use-case and requirement scope', and '3 Define trade study, variability axes and evaluation KPIs'.

The trade-study dashboard is the longest storyline: 150% architecture, feature model, mappings, configurations, 100% derivations, simulations, comparison and decision.

What scope changes do and do not do

- Changing scope does not delete model, variability, simulation or trade-study records.
- Changing scope changes the required dashboard steps and their completion logic.
- A blocked step explains the missing prerequisite. Repair the evidence; do not search for a manual completion switch.
- A green dashboard step means its implemented evidence gate is satisfied. It does not claim detailed-design verification.
- All analyses are preliminary engineering estimates and should carry assumptions and provenance.

Part III - Model workspace

Model is the canonical MBSE authoring workspace. Every row is part of the reusable 150% Product-Line Model. Configurations later include, exclude or modify that source through variation points.

Model tabs and the content visible in each

Tab	Editable element types	Available views
Mission and Context	Mission, stakeholder, need, objective, use case	Editable table, context graph, traceability matrix, Model quality
Requirements and Validation	System requirement, verification method	Editable table, trace diagram, traceability matrix, Model quality
Product Functional	Product function, product interface	Editable table, architecture graph, function sequence, traceability matrix, Model quality
Product Technical	Product component, product interface	Editable table, architecture graph, traceability matrix, Model quality
Process Functional	Process function, process interface	Editable table, architecture graph, function sequence, traceability matrix, Model quality
Process Technical	Industrial-system component, process interface, resource	Editable table, architecture graph, traceability matrix, Model quality
Interfaces and Traceability	Every element type	Editable table, cross-domain graph, global traceability matrix, Model quality

Top authoring controls

Field or control	What to enter or select	Allowed values / rule	Actual effect
New element type	Select the type to create in the current tab.	Only types valid for the selected tab.	MODEL. Determines the type of the row created next.
Add row	Click once after selecting a type.	No text required before creation.	MODEL. Creates a common draft 150% element with a generated ID, default tag and any custom-column defaults.
Add column	Click to define a reusable custom attribute.	Definition fields are documented later in this chapter.	MODEL/GOV. Adds the attribute to every element of one type.
Units	Click to inspect built-in units or define a project unit.	Built-in catalog plus validated custom definitions.	RUN. Unit definitions support formula compatibility and conversion.

Filters and context controls

Field or control	What to enter or select	Allowed values / rule	Actual effect
Search	Type any part of an element name, description, tag, parameter name, value or unit.	Free text; blank shows all.	VIEW only.
Element type	Choose All types or one type valid in the tab.	Tab-specific list.	VIEW only; also determines which type's table sort selector is shown.
Status	Choose All statuses, draft, reviewed or approved.	Four display choices.	VIEW only.
Model view	Choose 150% or a configuration / architecture.	150% plus non-archived architectures.	VIEW and calculation context; source remains unchanged.
Scope filter	Choose All scopes, common or specific.	Three choices.	VIEW only. New rows are common; configuration effects are modeled through variation points.
Tag filter	Type a tag fragment.	Free text; case-insensitive partial match.	VIEW only.
Clear task	Click to remove the dashboard workflow banner.	Only shown after dashboard navigation.	VIEW only.
Table order	Choose Manual order, Alphabetical A-Z or Alphabetical Z-A.	Three choices.	VIEW order. Dragging rows is enabled only in Manual order.
Graph layout	Choose Manual positions, Automatic hierarchy or Sequence stages when offered.	Choices depend on tab.	VIEW only; saved layout preference does not change model semantics.

Editable table columns

Field or control	What to enter or select	Allowed values / rule	Actual effect
Drag handle	Drag a row above or below another row of the same type.	Manual order only.	VIEW order stored per element type.
Name	Type a concise unique human-readable label.	Free text; non-empty is required by validation.	MODEL. Used throughout graphs, matrices, search and reports.
Type	Read the colored element-type badge.	Read-only.	READ. Type controls valid relations and tab placement.
Working scope	For a use case, check In working scope.	Enabled only when the use case is linked to the system of interest through involvedIn.	MODEL. Controls semantic scope, requirements and trade-study eligibility.
Status	Select draft, reviewed or approved.	Three values.	GOV. Affects maturity/validation; it does not change a numeric formula.
Authoritative duration	For a process function, type a non-negative number and choose minute, hour or day.	Other element types show a dash.	RUN. Feeds lead time, process cost, resource demand, utilization and throughput proxy.
Parameters and recommended ranges	Read parameter value/unit; optionally type minimum, maximum and uncertainty percent.	Numeric fields; uncertainty 0-100.	RUN/GOV. Bounds document expected range and may support uncertainty review; value is edited in full details.
Custom column	Enter text/number or choose True/False/Not set according to its definition.	Definition-specific.	MODEL/GOV. Can be required and varied; never a KPI formula token.
Tags	Type comma-separated tags.	Any non-empty tag strings.	DOC/VIEW. Search and filtering; tags can also be a variation target. Tags are not direct KPI inputs.
Connections	Read the count of incoming and outgoing relationships.	Read-only integer.	READ. It does not prove that the correct relation exists.
Edit / Open full details	Click to open the side editor.	Any row.	VIEW navigation.
Duplicate	Click to copy the element with a new ID.	Any row.	MODEL. Connections are not silently duplicated as equivalent semantics.
Delete	Click and confirm.	Any row; referencing relationships are also removed.	MODEL. Destructive; revalidate formulas and traceability afterward.

Element types - what each one represents

Element type	Meaning	Coffee example	Key storyline role
Mission	Outcome and context that justify the work.	Deliver a configurable coffee experience	Root of stakeholder intent.
Stakeholder	Person, organization or external system with a concern. Exactly one may be the system of interest.	Coffee Machine System; Coffee Consumer	Defines scope and owners of needs/objectives/use cases.
Need	Stakeholder-valued outcome, usually qualitative.	Fast beverage preparation	Derives mandatory requirements.
Objective	Measurable desired outcome.	Serve peak household demand	Links KPIs to decision value and derives requirements.
Use case	Observable interaction or service in the investigation.	Manufacture coffee machine	Working-scope gate for functions and trade study.
System requirement	Testable statement with optional executable comparison.	The machine shall remain below the mass limit	Feasibility gate before trade scoring.
Product function	Behavior performed by the delivered product.	Heat and pressurize brew water	Must be realized by product structure.
Product component	Physical or logical part of the delivered product.	Brewing and heating module	Owns product parameters and realizes product functions.
Product interface	Boundary or connection in the product architecture.	Thermal and hydraulic interface	Documents/relates product connectivity.
Process function	Manufacturing or verification activity.	Integrate coffee-machine modules	Owns authoritative duration and is realized by industrial equipment.
Industrial-system component	Equipment, workstation, cell, software or facility that realizes process behavior.	Coffee-machine integration cell	Owns industrial parameters and requires resources.
Process interface	Material, control or information connection in the industrial architecture.	Test-data interface	Documents/relates industrial connectivity.
Resource	Person, role, skill, tool, machine, software or facility required by an industrial component.	Manufacturing technician	Hourly rate and assignment quantity feed industrial calculations.
Verification method	Method used to verify a requirement.	End-of-line functional test	Connects verification intent to requirements and process activity.

Element full-details editor

Click Open full details or Edit. Fields above Save element are held in an editor draft until you click Save element. Parameter edits below that button are written immediately.

Field or control	What to enter or select	Allowed values / rule	Actual effect
Name	Type the label used by people and diagrams.	Required non-empty text.	MODEL. Rename-safe formula references remain stable because IDs do not change.
Description	Explain purpose, boundary and interpretation.	Free text; blank allowed.	DOC. Searchable and exported; not a numeric input.
Status	Select draft, reviewed or approved.	Three values.	GOV. Maturity and validation only.
Canonical 150% notice	Read only.	No value.	READ. Reminds the user that variability belongs in variation points.
Designate system of interest	For a stakeholder, check exactly one system that the project designs or assesses.	Boolean; stakeholder only.	MODEL. Critical semantic-scope and dashboard gate.
Tags	Type comma-separated classification/search labels.	Free strings; existing tags are suggested.	DOC/VIEW and possible variation target. Not a KPI input.
Owner	Type accountable role or person.	Free text; blank allowed.	GOV/DOC. Exported audit context; no calculation effect.
Source	Type the document or origin for the element definition.	Free text; blank allowed.	GOV/DOC. Different from parameter provenance; no direct calculation effect.
Process duration	For a process function, type its authoritative duration.	Non-negative number; complete evidence should be positive.	RUN.
Duration unit	Choose minute, hour or day.	Three choices.	RUN. One engineering day is interpreted as eight hours.
Resource type	For a resource, classify it.	person, role, skill, tool, machine, software, facility	MODEL/DOC classification; no algorithm branches by this value today.
Hourly rate	For a resource, enter its cost rate.	Number at least zero.	RUN. Used by processCost when the resource is required by a realizing industrial component.
Custom attributes	Enter values defined by Add column.	String, number or boolean per definition.	MODEL/GOV. Required values affect validation; values may be varied; not available as KPI formula tokens.
Save element	Click after changing common fields or a requirement formula.	Button action.	MODEL/GOV. Persists the editor draft and increments relevant revision state.
Connected elements	Click a connection to open its counterpart and select the relation.	Read-only list.	VIEW navigation.
Duplicate	Copy this element as a new model element.	Button action.	MODEL.
Delete	Confirm deletion; referencing relationships are removed.	Button action with confirmation.	MODEL, destructive.

Parameter card fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Add	Click to add New parameter to this element.	New parameters are numeric with blank value, origin entered and all configurations.	RUN. Creates a parameter with generated immutable ID.
Parameter name	Type a clear engineering quantity label.	Free text; changing it automatically regenerates the semantic key in this editor.	MODEL/RUN. Friendly name; KPI exact ID remains stable.
Value origin	Choose entered, assumed, calculated or simulated. Calculated appears only on product/industrial components.	Four values subject to element type.	GOV/RUN. Calculated opens a formula and makes value/unit read-only.
Value	Type the engineering value or leave blank when unavailable.	Numeric in the current user-created parameter path; blank becomes null.	RUN. Direct formula, requirement and KPI evidence.
Unit	Type a built-in or project unit symbol.	Compatible unit string; blank allowed but weakens evidence.	RUN. Formula compatibility and output conversion.
Source / provenance	Identify document, supplier, assumption, calculation or simulation source.	Free text.	GOV. Retained with simulation input evidence; does not change the number.
Minimum	Enter recommended lower bound.	Finite number or blank.	GOV/RUN support. Used as uncertainty/range evidence, not as the nominal value.
Maximum	Enter recommended upper bound.	Finite number or blank; should be at least minimum.	GOV/RUN support.
Uncertainty +/- %	Enter symmetric uncertainty around the nominal value.	0 to 100 or blank.	GOV/RUN support for uncertainty review; base simulation uses nominal value.
Applicable configurations	Check only configurations that use this parameter value. Leave all unchecked for every configuration.	Any active configurations; empty means all.	RUN. Determines which value appears in each configured calculation.
Delete parameter	Click the red trash button.	No inline dependency preview here.	RUN/MODEL. Can invalidate requirement and KPI bindings.

Parameter identity

The stable parameter ID, not its displayed name, is the canonical formula reference. Renaming is safe. Deleting and recreating a parameter is not equivalent; rebind formulas after replacement.

Calculated parameter formulas

Calculated parameters are supported only on Product component and Industrial-system component. They may read parameters on the same component and on components that transitively refine it.

1	Create the target Open the component, click Add under Parameters, and rename New parameter to Volume. Why: The target owns the calculated value. Done when: A Volume parameter card exists.
2	Choose calculated Open Value origin and select calculated. Why: This opens the formula editor and makes target value/unit read-only. Done when: Calculated parameter formula is visible.
3	Insert the first source Open Select hierarchy parameter, choose Length, and click Insert. The editor appends a symbol such as @length and stores the selected parameter ID. Why: Using the picker creates an immutable binding. Done when: A binding row for @length is shown.
4	Insert other sources Repeat for Width and Height. Between the inserted symbols type * so the expression reads @length * @width * @height. Why: Multiplication preserves unit dimensions. Done when: The expression contains three bound symbols.
5	Request an output unit In Requested output unit type m^3 only if the source unit catalog can convert to it; otherwise leave blank for inferred unit. Why: The requested unit changes the displayed calculated unit, not the physical quantity. Done when: Status becomes calculated or gives a specific unit error.
6	Read status Inspect calculated, pending or error and its message. Repair missing values, bindings, syntax or units before using the result. Why: Downstream requirements and KPIs need a numeric current value. Done when: Volume displays a value and unit.

Calculated-parameter grammar

Construct	Example	Rule
Bound reference	@length	Must have a binding row created by Insert.
Literal number	2.5	Dimensionless numeric literal.
Arithmetic	@a + @b; @a - @b; @a * @b; @a / @b	Addition/subtraction require compatible dimensions; multiplication/division compose dimensions.
Parentheses	(@a + @b) / 2	Control precedence.
Square root	sqrt(@area)	Argument must be non-negative and dimension exponents must remain valid.
Power	@length ^ 2 or @length ** 2	Exponent syntax supports ^ and **.
Not supported	JavaScript, display-name lookup, arbitrary functions	The safe parser accepts only the documented grammar.

Requirement formulas - complete procedure

A requirement formula evaluates one comparison. A symbol is readable text; its binding stores an immutable parameter or KPI ID. Manually typed symbols without bindings remain pending and count as failed evidence.

Open a System requirement to enter its expression, insert immutable parameter/KPI bindings and inspect satisfied, failed, pending or error status.

1 Open the requirement

Go to Model > Requirements and Validation > Editable table. Click Open full details for The machine shall remain below the mass limit.

Why: The formula is stored on the requirement.

Done when: Requirement formula is visible in the side editor.

2	Select the evidence Open Select parameter or future KPI. Choose Brewing and heating module - Mass (kg). Click Insert. Why: The picker stores the exact parameter ID and generates a readable @symbol. Done when: A binding row such as @equipment_mass appears.
3	Complete the comparison In the formula box type the comparator and threshold after the inserted symbol so the expression is @equipment_mass <= 1000. Why: Exactly one comparison turns arithmetic into pass/fail evidence. Done when: The editor shows satisfied or failed if the value is available.
4	Save the requirement Click Save element. Do not close the panel before saving. Why: Requirement-formula changes are part of the element draft. Done when: The saved formula remains after reopening the requirement.
5	Change evidence safely Edit the source parameter value, not the symbol text. Return to the requirement and read the new status. Why: The binding follows the stable parameter ID. Done when: Actual status updates from the current value.
6	Repair pending symbols If a symbol was typed manually, remove it or use the picker to create a binding and then use the inserted symbol. Remove obsolete bindings with X. Why: Text alone does not identify a model value. Done when: Every @symbol in the expression has a visible binding row.

Requirement formula fields and grammar

Field or control	What to enter or select	Allowed values / rule	Actual effect
Expression	Type arithmetic followed by exactly one comparison.	+, -, *, /, parentheses; comparator <, <=, >, >=, =, == or !=.	RUN. Produces requirement status and trade-study feasibility.
Reference picker	Choose any model parameter or a KPI definition.	Parameters show owner/name/unit; KPIs show output unit.	RUN. Creates an immutable binding.
Insert	Click after choosing a reference.	Adds a unique @symbol; suffixes such as _2 avoid duplicates.	RUN.
Binding row	Read symbol and source. Click X only when removing that binding.	parameter or KPI ID behind the symbol.	READ/RUN. Missing binding makes the expression pending.
Evaluation status	Read satisfied, failed, pending, error or notDefined.	Read-only.	READ. Pending/error does not establish mandatory feasibility.

Example	Meaning	Important unit rule
@mass <= 1000	Mass may not exceed 1000 in the parameter's compatible unit context.	A bare threshold is interpreted against the bound quantity; use compatible units.
@length + @margin <= 800	Total length and margin must fit an envelope.	Addition requires compatible dimensions.
@cost_a + @cost_b < @budget	Combined cost must remain below a bound parameter/KPI.	All compared quantities must have compatible dimensions.
@throughput >= 10	Throughput must be at least ten.	Bound throughput unit and threshold interpretation must be explicit in the requirement text.
@quality != 0	Quality evidence must be nonzero.	Equality uses numeric tolerance.

Formula limitation

Requirement formulas are numeric. Textual verification rationale belongs in the requirement description and Verification method trace, not in the expression.

Custom columns and project units

Add reusable custom-attribute column

Field or control	What to enter or select	Allowed values / rule	Actual effect
Element type	Choose which element type receives the column.	Any model element type.	MODEL. The definition applies only to this type.
Column name	Type the label shown in the table and details editor.	Required non-empty text.	MODEL/DOC.
Data type	Choose string, number or boolean.	Three choices.	MODEL. Controls editor and validation.
Unit	For number only, type an engineering unit.	Optional; disabled for string/boolean.	DOC/GOV. It does not make the custom value a KPI parameter.
Default value	Type the value assigned when future elements of this type are created.	Parsed according to data type.	MODEL. Existing elements expose the field but are not silently reinterpreted as formula parameters.
Required	Check when every element of the type must have a value.	Boolean.	GOV. Missing values appear in validation/maturity.
Description	Explain intended meaning and allowed vocabulary.	Free text.	DOC only.
Create	Click after the name is entered.	Button enabled with non-empty name.	MODEL.

Custom-column impact

A custom attribute is stored, edited, exported, validated when required and can be targeted by a variation point. It is not offered in requirement/KPI parameter pickers and does not feed standard algorithms.

Engineering unit catalogue

Field or control	What to enter or select	Allowed values / rule	Actual effect
Built-in catalogue	Inspect symbol, quantity and reference.	Read-only built-in definitions.	READ/RUN support.
Symbol	Type the short notation used on parameters, for example ft.	Required unique text.	RUN.
Name	Type the full unit name, for example foot.	Required text.	DOC/RUN.
Quantity name	Type the physical quantity, for example length.	Required text.	RUN. Documents dimensional meaning.
Dimension exponents	Type a JSON object such as {"length":1} or {"length":1,"time":-1}.	JSON object with finite numeric exponents.	RUN. Determines unit compatibility.
Factor to SI	Type the multiplier from this unit to its SI representation.	Finite positive number.	RUN. Used for conversion.
Aliases	Type alternative symbols separated by commas.	Optional strings.	RUN/VIEW. Recognized alternative spelling.
Add custom unit	Click after completing valid fields.	Validation rejects duplicates/invalid dimensions/factors.	RUN.
Delete project unit	Click trash next to a project-defined unit.	Built-in units cannot be deleted.	RUN. Existing formulas/parameters may need repair.

Relationships and traceability

Relationships carry the semantic thread. Source, relation and target must match an allowed canonical triple. The UI filters target choices to compatible element types.

Canonical relation	Meaning	Valid source -> target	Calculation or storyline effect
hasStakeholder	Mission includes a stakeholder.	Mission -> Stakeholder	MODEL. Dashboard intent gate.
hasNeed	Stakeholder owns or expresses a need.	Stakeholder -> Need	MODEL. Scope and requirement derivation context.
hasObjective	Stakeholder owns a measurable objective.	Stakeholder -> Objective	MODEL. KPI eligibility and trade-study scope.
involvedIn	Stakeholder participates in a use case.	Stakeholder -> Use case	MODEL. System-of-interest eligibility and working scope.
hasFunction	Use case requires a function.	Use case -> Product function or Process function	MODEL. Dashboard architecture-completeness gate.
derives	Intent derives a requirement.	Need/Objective -> System requirement	MODEL/RUN. Mandatory requirement scope.
satisfiedBy	Architecture evidence satisfies a requirement.	Requirement -> Product/process function or Product/industrial component	MODEL. Traceability and parameter-binding helper.
refines	Child decomposes a parent.	Requirement->requirement; product function/component->same type; process function/industrial component->same type	MODEL. Hierarchy; calculated parameters may read transitive component refiners.
verifies	Verification method verifies a requirement.	Verification method -> Requirement	MODEL/GOV.
realizedBy	Function is implemented by technical structure.	Product function -> Product component; Process function -> Industrial component	MODEL/RUN. Core architecture gate; process algorithms use realization chain.
connects	Element connects to an interface.	Product function/component/interface -> Product interface; Process function/industrial component/interface -> Process interface	MODEL. Connectivity/docu mentation.
precedes	Function immediately precedes another in one sequence.	Product function -> Product function; Process function -> Process function	RUN. Process critical path and handoff validation.
allocatedTo	Function/method is allocated to product or process context.	Process function -> Product component; Resource -> Industrial component; Verification method -> Process function	MODEL. Product/process or verification allocation; resources for algorithms still require requiresResource from component.
requiresResource	Industrial component requires a resource quantity.	Industrial component -> Resource	RUN. Process cost, resource demand and utilization.
consumes	Process function consumes a product component item flow.	Process function -> Product component	MODEL/RUN. Requires positive quantity and unit; validates handoff.

Canonical relation	Meaning	Valid source -> target	Calculation or storyline effect
produces	Process function produces a product component item flow.	Process function -> Product component	MODEL/RUN. Requires positive quantity and unit; validates handoff.

Relationship editor fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Source	Select the canonical source element.	Any model element; later choices are filtered.	MODEL.
Relationship	Select a relation allowed from the source type.	Compatible relation list only.	MODEL/RUN depending on type.
Target	Select a compatible target.	Filtered element list; source cannot equal target.	MODEL.
Quantity	For requiresResource, consumes or produces, enter a positive quantity.	Finite number > 0; disabled otherwise.	RUN.
Unit	For quantity relations, type the unit; mandatory for item flows.	Non-empty for consumes/produces.	RUN/MODEL.
Item flow	For consumes/produces, optionally name the material/product flow.	Optional text.	DOC. Quantity/unit carry enforced semantics.
Create	Click to validate and save the new relation.	All required choices must be compatible.	MODEL.
Existing Name	Optionally type a relationship label.	Free text.	DOC only.
Existing Description	Optionally explain assumptions or boundary.	Free text.	DOC only.
Open	Select relation and its source context.	Existing row.	VIEW.
Save icon	Click after editing an existing relationship row.	Existing row.	MODEL/RUN.
Delete icon	Click and confirm.	Existing row.	MODEL/RUN, destructive.

Traceability matrix

Save Project Load Project 150% Product-Line Model

Local autosave: Saved Revision 1 Reset sample Reset all

Build the mission-to-validation thread in any order; the workflow and validation remain live. Mission + Add row + Add column Units

Mission and Context Requirements and Validation Product Functional Product Technical Process Functional Process Technical Interfaces and Traceability

Search elements, tags, parameters... All types All statuses

150% Product-Line Model All scopes Filter tag

Editable table Context graph Traceability matrix Model quality Manual order 13 of 41 elements

Name	Type	Working scope	Status	Authority
Deliver a configurable coffee experience Open full details	Mission	—	reviewed	—
Coffee Machine System Open full details	Stakeholder	—	reviewed	—
Product Manager Open full details	Stakeholder	—	reviewed	—
Coffee Consumer Open full details	Stakeholder	—	reviewed	—
Service and Quality Manager Open full details	Stakeholder	—	reviewed	—

The matrix keeps selected and previous-step elements visible. Horizontal scrolling is intentional; use the sticky first column and Link actions.

1

Choose the right tab

Open the domain tab, then click Traceability matrix. Use Interfaces and Traceability > Global traceability for the full cross-domain matrix.

Why: Column types are determined by the selected model section.

Done when: Selected element is the sticky first column.

2

Create a link

Find a row, scroll to the target type column, and click + Link.

Why: The dialog only offers compatible targets and canonical directions.

Done when: Create traceability link opens.

3

Choose target or nested parameter

Select an element. For a system requirement and an eligible architecture type, you may choose a nested parameter instead.

Why: Nested-parameter choice automatically creates/reuses satisfiedBy and a formula binding.

Done when: The dialog shows either relation choices or binding explanation.

4

Choose canonical relation

Select the relation direction shown by the full source -> relation -> target label, then click Create relationship.

Why: Reverse-facing cells still store canonical direction.

Done when: A blue relationship card appears in the cell.

5

Remove a wrong trace

Click the trash icon on the blue card and confirm.

Why: Deletion removes the canonical relationship everywhere.

Done when: Tables, graphs and validation update.

Graphs, sequences and Model quality

Field or control	What to enter or select	Allowed values / rule	Actual effect
Graph node	Click to select an element; use the side editor to modify details.	Current projection only.	VIEW navigation.
Graph layout	Manual positions, Automatic hierarchy, or Sequence stages where available.	Tab-specific.	VIEW only.
Named sequence	Select an existing function sequence.	Sequences of the current product or process domain.	MODEL/RUN context.
Sequence name	Type a clear scenario or behavior name.	Free text.	DOC/MODEL.
Use case described	Select one eligible working-scope use case involving the system of interest.	Exactly one per sequence.	MODEL. Changing it can clear selected functions and precedence connectors.
Architecture	Read Common / all or architecture context.	Read-only.	READ.
Sequence description	Explain scenario and boundary.	Free text.	DOC only.
Related function hierarchy	Check every related leaf function needed in the scenario.	Parent functions are context-only and cannot be selected.	MODEL/RUN. Sequence completeness and lead-time graph.
Predecessor / Successor	Choose two selected leaf functions and click Link.	No self-link; graph must remain valid/acyclic for lead time.	RUN. Creates precedes in this sequence.
Precedence description	Type handoff notes in the synchronized table.	Free text.	DOC only.
Process flow Direction	Choose Consumes or Produces.	Two values.	MODEL/RUN item-flow semantics.
Process flow Product component	Choose the consumed/produced product component.	Active compatible product component.	MODEL/RUN.
Process flow Quantity / Unit	Enter positive quantity and a non-empty unit.	Quantity > 0; unit required.	RUN/MODEL handoff validation.
Item-flow name	Optionally name the handoff.	Free text.	DOC only.
Validation severity	Filter All, error, warning or information.	Four display choices.	VIEW only.
Validation category	Filter current categories such as identity, relationship, traceability, completeness, process, parameter, feature, formula, comparison, export or persistence.	Only categories present in current results are listed.	VIEW only.
Validation workspace/domain	Filter All, Model, Variability, Parameters and KPIs, or Simulation.	Five display choices.	VIEW only.
Run validation	Click to recompute deterministic complete-project checks.	Button action.	GOV/MODEL evidence. Does not fix findings automatically.
Finding	Click a finding to open one affected object or choose among several.	Read-only result.	VIEW navigation.

Sequence semantics

The sequence uses immediate precedence. Parallel splits and joins use AND semantics. Manufacturing lead time follows the longest active path. Item-flow handoff checks compare produced and consumed product components plus units.

Coffee-machine model inspection

1 Inspect intent

Open Mission and Context > Traceability matrix. Click Deliver a configurable coffee experience and follow hasStakeholder to Coffee Machine System, Product Manager, Coffee Consumer and Service and Quality Manager.

Why: This demonstrates the justification chain.

Done when: Mission/stakeholder cards are visible.

2 Inspect scope

Open the Coffee Machine System stakeholder. Confirm Designate this stakeholder as the system of interest is checked. In the table, confirm the three coffee use cases are eligible and selected as intended.

Why: Only use cases involving the system of interest can be in working scope.

Done when: Working scope has no red Not linked message.

3 Inspect product realization

Open Product Functional > Traceability matrix. Follow Meter coffee ingredients, Heat and pressurize brew water and Monitor beverage quality to their realizing product components.

Why: Every product function must have technical realization.

Done when: Each function has realizedBy evidence.

4 Inspect industrial realization

Open Process Technical > Architecture & hierarchy. Follow Manufacture coffee-machine modules, Integrate coffee-machine modules and Test completed coffee machine to the workstation, integration cell and test station.

Why: Every process function must have industrial realization.

Done when: Each process function has realizedBy evidence.

5 Inspect resource rule

In Process Technical > Editable table, scroll to Relationship editor. Confirm requiresResource starts at an industrial component and targets Manufacturing technician, End-of-line test technician, Automated module assembly cell or Automated component feeder.

Why: This is the authoritative allocation path.

Done when: No resource is attached directly to a function through requiresResource.

Part IV - Variability and 150% to 100% transformation

Variability connects selectable product-line choices to model effects. The canonical 150% model remains unchanged. A configuration stores feature selections; a derivation stores one immutable 100% result from one source revision.

Variability menu

Tab	Purpose	Editable?	Main output
Feature Model	Define selectable features, organization and cross-tree constraints.	Yes	One valid shared feature model.
Variation Points	Map feature conditions to element/relationship existence or property effects.	Yes	Deterministic 150% transformation rules.
Configurator	Create named feature selections and validate them.	Yes	Valid configuration plus linked architecture identity.
150% Preview	Preview a configuration overlay on the editable family model.	Source is editable; overlay is computed	Included/excluded/modified visual forecast.
100% Realization	Inspect a saved immutable derived architecture.	Read-only	Traceable 100% architecture and applied-variation audit.
Derivation Summary	Inspect selection, scope, counts, revision, timestamp and KPI snapshot.	Read-only	Compact derivation evidence.
Variability validation	Open Model quality filtered to variability.	Read-only findings; repairs occur in editors	Feature/mapping/configuration errors.

Feature Model main controls

Action	What happens	Use it when	Impact
Graph	Shows feature hierarchy, groups and constraints graphically.	Understanding structure or selecting a feature.	VIEW.
Table	Shows feature fields, stable IDs and actions.	Searching, editing, duplicating or deleting.	VIEW and MODEL through row actions.
Group	Creates a non-selectable organizational FeatureGroup.	Organizing major design axes.	MODEL.
Constraint	Creates requires or excludes between features.	Rules not expressible by hierarchy/cardinality alone.	MODEL/RUN.
Feature	Creates a selectable or structural feature.	Adding variability choices.	MODEL/RUN.
Search features	Filters the table by name or ID.	Locating a feature.	VIEW only.
Duplicate feature	Copies a non-root feature with new identity.	Creating similar choices.	MODEL. Review parent/group/constraints afterward.
Delete feature	Deletes feature references and affected descendants/constraints/config selections/variation points after confirmation.	Removing obsolete variability deliberately.	MODEL/RUN, destructive.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saving

Revision 1

Reset sample

Reset all

Variability

Variability validation

One canonical feature model drives validated selections and non-mutating 150% → 100% derivation.

FeatureModel

Variation Points

Configurator

150% Preview

100% Realization

Derivation Summary

Feature model

Containment is neutral; requires is green and directional; excludes is red. Organizational groups are recursive and nonselectable.

Graph

Table

+ Group

Constraint

+ Feature

Feature hierarchy

Containment defines the tree; cross-tree constraints are available on demand.

LAYOUT

Automatic hierarc

Requires / excludes

Auto-arrange

Feature Model shows the shared root and selectable coffee-machine design choices used by every configuration.

Feature editor fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Name	Type the human-readable choice.	Required non-empty text.	MODEL. Used by configurator and reports; expressions still reference stable ID.
Description	Explain what selecting the feature means.	Free text.	DOC only.
Cardinality / feature type	Choose root, mandatory, optional, xor or or.	Exactly one value.	RUN/MODEL. Controls automatic/allowed selection.
Parent feature	For every non-root feature, choose its parent.	Required except root; cannot select itself.	MODEL. Defines hierarchy.
Cardinality group ID	For xor/or members, type a shared group identifier for siblings in the same choice set.	Required for xor and or.	RUN. XOR permits exactly one effective member; OR permits one or more.
Organizational group	Optionally place the feature in a named FeatureGroup.	No group or any existing group.	MODEL/DOC organization; trade-study axes link to groups.
Value type	Choose boolean or enumeration.	Two values.	RUN. Boolean is selected/unselected; enumeration also stores a typed value.
Variability classification	Choose external or internal.	Two values.	MODEL/DOC classification. Both can affect derivation; internal is commonly implementation/applicability content.
Allowed values	For enumeration, type comma-separated values.	At least one distinct value.	RUN. Configurator restricts selection to these values.
Default value	For enumeration, select one allowed value.	Any allowed value.	RUN/HINT. Configurator suggests it but the user must still create a valid effective value.
Sibling sort order	Enter a non-negative order number.	Integer-like number ≥ 0 .	VIEW order only.
Save	Click after completing required fields.	Side-editor action.	MODEL/RUN.

Feature type behavior

Type	Configurator behavior	Use it for	Coffee example
root	Always effective; exactly one project root is required.	The full product-line identity.	Coffee Machine Product Line
mandatory	Automatically selected when its parent is effective.	Content every relevant variant needs.	Common Brew Control; Food and Electrical Safety
optional	User may select or clear it.	Independent capability.	Brew History; Adaptive Brew Profile
xor	Exactly one member of the same group is effective.	Mutually exclusive design alternatives.	Capsule Input versus Automatic Bean Grinder
or	One or more members of the same group may be effective.	Combinable alternatives with minimum-one semantics.	Beverage Sensing and Milk Temperature Probe

Organizational group fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Name	Type the grouping/axis label.	Required text.	MODEL/DOC. Trade-study variability axis can own this group.
Description	Explain the design decision represented.	Free text.	DOC only.
Parent feature	Choose a feature to contain the group.	A parent feature or, if blank, a parent group.	MODEL hierarchy.
Parent organizational group	When no parent feature is chosen, nest under another group.	Existing group except itself.	MODEL hierarchy.
Sort order	Enter non-negative display order.	Number ≥ 0 .	VIEW only.

Feature constraint fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Source feature	Choose the feature whose selection triggers the rule.	Any feature.	RUN/MODEL.
Relationship type	Choose requires or excludes.	Two values.	RUN. requires is directional; excludes blocks both together.
Target feature	Choose the other feature.	Any feature except the source.	RUN/MODEL.

Constraint behavior

When a requires dependency is encountered, the configurator can add it automatically and records it as automatic rather than manual. An excludes conflict makes the configuration invalid until one conflicting choice is removed.

Variation Points

A variation point is the explicit bridge between feature selection and physical/functional architecture content. It can target elements or relationships. Do not encode configuration behavior only in descriptions.

Variability

One canonical feature model drives validated selections and non-mutating 150% → 100% derivation.

Feature Model **Variation Points** Configurator 150% Preview 100% Realization Derivation Summary

Graph List + Variation point

Modeling sections

All ☐ Mission and Context ☐ Requirements and Validation

☐ Product Functional ☐ Product Technical ☐ Process Functional

☐ Process Technical ☐ Interfaces and Traceability

Element types

All ☐ Mission ☐ Stakeholder ☐ Need ☐ Objective

☐ Use case ☐ System requirement ☐ Product function

☐ Product component ☐ Product interface ☐ Process function

41 element(s) and 74 relationship(s). A relationship is shown only when both endpoints are visible.

Cross-domain variation-point graph

Double-click an element body for Existence, a parameter or attribute for Property/Tag, or a connector for a relationship variation point. Existing mappings open for editing.

Readable graph view
Types occupy fixed top-down semantic bands.

RELATIONSHIPS CARD DETAIL

Primary structure Compact Auto-arrange

Variation Points list their kind, scope, feature condition, targets and ordered effects; disabled mappings remain stored but do not apply.

Variation Points main controls

Action	What happens	Use it when	Impact
Graph	Shows the canonical model filtered by modeling sections and element types.	Creating/editing mappings directly from graph targets.	VIEW plus MODEL when double-click creates a mapping.
List	Shows mapping name, kind, scope, enabled state, condition, targets and effect summary.	Auditing and editing mappings.	VIEW.
Variation point	Creates a blank mapping.	Adding a rule deliberately.	MODEL/RUN.
Modeling sections	Check domain sections or All.	Any combination.	VIEW only.
Element types	Check target element types or All.	Any combination.	VIEW only.
Edit	Opens the full variation-point editor.	Existing mapping.	VIEW navigation.
Disable / Enable	Toggles whether the mapping participates.	Boolean.	RUN. Disabled mappings are ignored by preview/derivation.
Delete	Deletes the mapping after confirmation.	Existing mapping.	RUN/MODEL, destructive.

Variation-point fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Name	Type a result-oriented mapping name.	Required non-empty text.	MODEL/DOC.
Description	Explain why and how the effect applies.	Free text.	DOC only.
Kind	Choose Existence, Primitive Property, Primitive Tag or Element Property.	Four values.	RUN. Determines whether content is removed or a safe property is modified.
Target type	Choose element or relationship.	Two values.	RUN/MODEL.
Targets	Check one or more canonical targets.	At least one required.	RUN. The same mapping effect is evaluated for each selected target.
ImpactedByFeature	Build a boolean feature-ID expression.	Blank = always active; feature IDs with AND, OR, NOT and parentheses.	RUN. Top-level applicability of this mapping.
Feature search	Search feature name, ID or group path, then click a result to insert its ID.	Any feature.	VIEW/RUN. Search is view; inserted ID is execution input.
Impacted feature values	Add enumeration conditions, then choose feature, equals/not equals, and an allowed value.	Enumeration features only.	RUN. Adds typed-value gating.
Realization domain	Choose Every scope or requirements, structure, behavior, process, resources or verification.	Seven choices.	RUN. Applies only when the configuration realization scope includes that domain.
Safe property path	For non-existence kinds, type an allowed path.	Examples: description, metadata:duration, parameter::value, tags, sourceId, targetId, architectureId.	RUN. Identifies the modified stored property.
Ordered value rules	Add one or more rules. First matching rule supplies the realized value; a blank condition is a fallback.	At least one for property/tag kinds.	RUN.
Rule feature expression	Build the feature-ID expression for that value.	Blank means fallback/always for this rule.	RUN.
Rule feature-value conditions	Add typed enumeration comparisons.	equals or not equals against allowed value.	RUN.
Realized value	Type the value assigned when the rule matches.	Number, text, true, false, null or JSON string array.	RUN.
Enabled	Check to make the mapping active.	Boolean.	RUN.

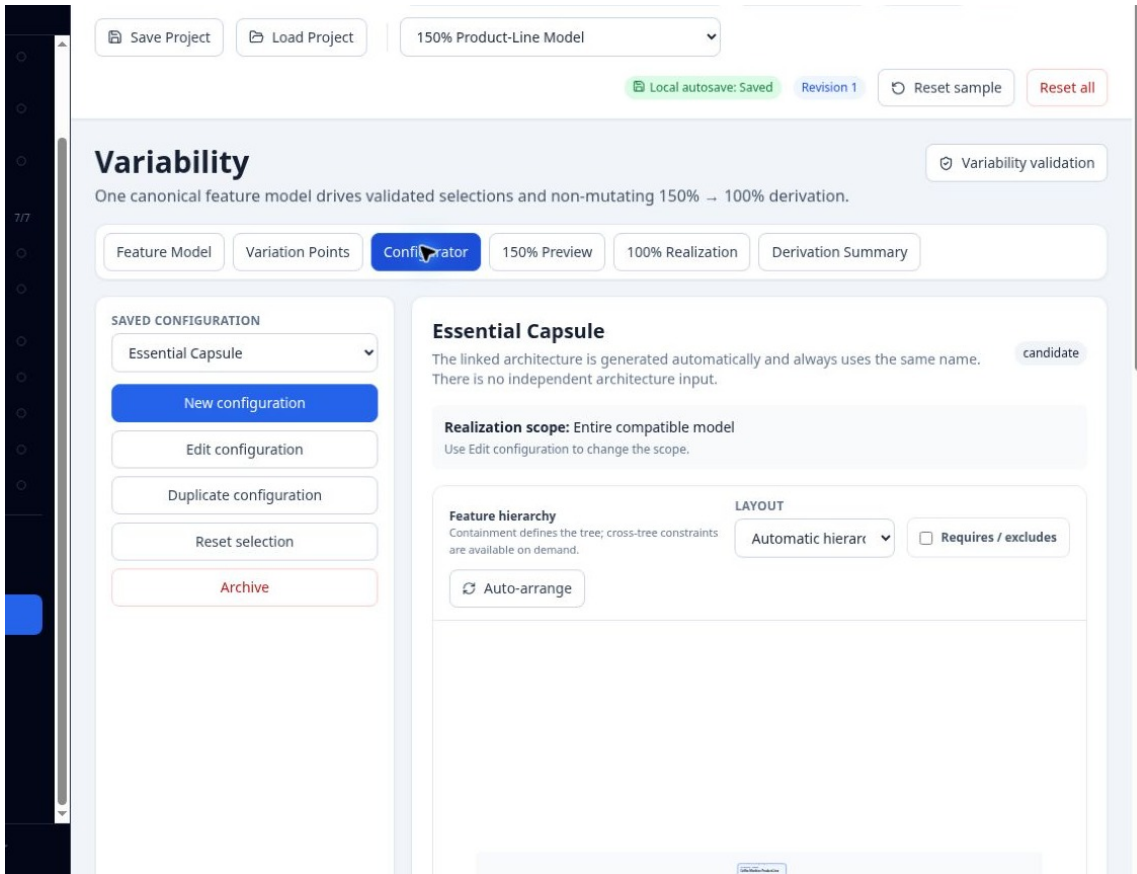
Variation-point kinds

Kind	Effect	Typical safe path	Coffee use
Existence	Keeps target when condition is true and removes it when false.	No property path	Include Automated component feeder only for Automatic Bean Grinder.
Primitive Property	Changes a primitive stored value.	metadata:duration; description; architectureId	Set manufacturing duration to 4 or 1.5 minutes by loading choice.
Primitive Tag	Changes tags with a string-array value.	tags	Apply variant classification tags.
Element Property	Changes an allowed element/relationship property, including a parameter value.	parameter::value; sourceId; targetId	Set variant-specific mass, cost or power input.

Feature-expression examples

Expression	Meaning	Result when blank
feature-automatic-grinder	Active only when that feature is effective.	Not applicable.
feature-dual-boiler AND feature-beverage-sensing	Both choices must be effective.	Not applicable.
feature-capsule OR feature-automatic-grinder	Either choice activates the mapping.	Not applicable.
feature-brew-history AND NOT feature-adaptive-profile	History selected and adaptive profile not selected.	Not applicable.
	No boolean feature gate.	Always active, subject to typed conditions and realization scope.

Configurator



Configurator stores a named selection, distinguishes manual from automatic features, validates rules and derives the linked 100% architecture.

Configuration editor fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Configuration name	Type the variant name.	Required non-empty text.	MODEL. The linked architecture is automatically generated with the same name.
Entire compatible model	Check for all compatible realization domains.	Boolean; represented by an empty stored scope list.	RUN. Applies all enabled variation domains.
Requirements / structure / behavior / process / resources / verification	Check only the domains to realize when not using Entire compatible model.	Any subset.	RUN. Variation points outside selected domains do not apply.

Configurator actions and feature rows

Field or control	What to enter or select	Allowed values / rule	Actual effect
Saved configuration	Choose an active configuration.	Any non-archived configuration.	VIEW/MODEL context.
New configuration	Click and complete the side editor.	Button action.	MODEL.
Edit configuration	Change name or realization scope.	Disabled for archived configuration.	MODEL/RUN. Resets validation to notValidated.
Duplicate configuration	Create an editable copy with no derivation/history.	Any selected configuration, including archived.	MODEL.
Reset selection	Clear manual/automatic selections and typed values.	Active configuration only.	RUN/MODEL.
Delete / Archive	Delete when no simulation history; archive when immutable history references it.	Confirmation required.	MODEL/GOV. Archived content is retained read-only.
Feature checkbox	Check or clear selectable features.	Root/mandatory are disabled and automatic; XOR clears siblings; constraints may add dependencies.	RUN.
Selection badge	Read Manual, Automatic, Invalid or Not selected.	Read-only.	READ.
Enumeration value	Choose one allowed value for an effective enumeration feature.	Allowed values from feature; empty is invalid.	RUN.
Validate	Click after editing selections or values.	Active configuration.	RUN/GOV. Sets valid/invalid and ordered findings.
Derive 100% model	Click only when no validation errors remain.	Active valid configuration.	RUN. Creates or refreshes immutable derivation.
Validation status	Read notValidated, valid or invalid.	Read-only.	READ.
Derivation status	Read None, Current or Stale.	Read-only.	READ. Stale means source/model basis changed.

Architecture identity

There is no independent architecture-name field. Creating or renaming a configuration creates or renames its linked architecture with the same name.

Coffee configuration procedure

1

Create the configuration

Click New configuration, type Essential Capsule, leave Entire compatible model checked, and save.

Why: This creates the configuration and linked architecture identity.

Done when: Essential Capsule appears in Saved configuration.

2

Select design features

Check Capsule Input and Single Thermoblock. Select the intended inspection/data options. Mandatory Common Brew Control and Food and Electrical Safety remain automatic.

Why: Manual selections state the external design choice.

Done when: Rows show Manual or Automatic.

3 Set Regional Voltage

If Regional Voltage is effective, choose 230V or 400V from its value list.

Why: Enumeration features need a typed value, not only a checkbox.

Done when: No invalid enumeration finding remains.

4 Validate

Click Validate. Read every ordered finding. Correct XOR/OR, requires/excludes and missing typed values, then validate again.

Why: Derivation must not bypass feature-model errors.

Done when: Badge shows valid.

5 Derive

Click Derive 100% model. If a current derivation already exists, use the explicit transform/rederive control when source revision changes.

Why: A derivation is tied to one model revision.

Done when: Derivation status shows Current.

6 Repeat alternatives

Duplicate and rename the configuration to Balanced Bean-to-Cup and Premium Dual Boiler, then change the feature selections, validate and derive each.

Why: Trade study needs at least two current comparable alternatives; the sample has three.

Done when: Each alternative has a Current derivation.

150% Preview, 100% Realization and Derivation Summary

Field or control	What to enter or select	Allowed values / rule	Actual effect
Model-view configuration overlay	Choose a configuration or 150% no-configuration view.	Active configurations only.	VIEW. Drives preview/realization/summary context.
Transform 150% -> 100%	Click to derive or rederive the selected configuration.	Selected active valid configuration.	RUN.
Preview status	Read common, included, excluded or modified.	Read-only visual classification.	READ.
Add Element	In 150% Preview, click Element and complete the new-element editor.	Any element type; process duration fields appear for process functions.	MODEL.
Common to every configuration	Keep checked for reusable common content; clear for selected-configuration-only content.	Shown when a configuration overlay is selected.	RUN/MODEL. Clearing automatically creates an internal applicability feature and existence variation point.
Show removed content as red audit overlay	Check in 100% Realization.	Boolean.	VIEW only; removed source content is not reinserted.
100% inspector	Click a derived graph item and read ID, status and parameters.	Read-only.	READ.
Applied variation audit	Read variation point, target, effect, path, previous and next value.	Read-only.	READ. Exact trace from 150% source to 100% result.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

Variability

One canonical feature model drives validated selections and non-mutating 150% → 100% derivation.

Feature Model

Variation Points

Configurator

150% Preview

100% Realization

Derivation Summary

MODEL-VIEW CONFIGURATION OVERLAY

Essential Capsule The canonical 150% model is always available. Generated configuration views are overlays or immutable realizations.

Read-only immutable 100% realization Current Transform 150% → 100% Show removed content as red audit overlay

Readable graph view

Types occupy fixed top-down semantic bands.

Auto-arrange

RELATIONSHIPS

Primary structure

CARD DETAIL

Compact

SYSTEM REQUIREMENT - READ-ONLY INSPECTOR

The machine shall remain below the mass limit

The machine shall remain below the mass limit in the coffee-machine product-line model.

ID: systemRequirement-the-system-shall-remain-below-the-equipment-mass-target

Status: included

Parameters: None

The 100% Realization is read-only and revision-aware. Its audit explains every applied existence/property effect.

New element in 150% Preview

Field or control	What to enter or select	Allowed values / rule	Actual effect
Element type	Choose the intended canonical type.	Any model element type.	MODEL.
Name	Type the element label.	Required non-empty text.	MODEL.
Description	Explain purpose/boundary.	Free text.	DOC.
Tags	Type comma-separated tags.	Optional; default type tag is used when blank.	DOC/VIEW/variation target.
Duration / unit	For a process function, type positive duration and select minute/hour/day.	Both or neither; duration > 0 when entered.	RUN.
Common to every configuration	Choose common versus current configuration-specific.	Boolean.	RUN/MODEL automatic mapping as described above.

Derivation Summary fields

Read-only field	Meaning	Why it matters
Manual selections	Choices explicitly made by the user.	Separates intent from constraint-driven selections.
Automatic selections	Features added by mandatory/constraint logic.	Shows hidden dependency effects.
Effective selections	Complete feature set used by derivation.	Canonical variant definition.
Typed feature values	Enumeration values used.	Explains property effects such as Regional Voltage.
Realization scope	Domains included in transformation.	Explains why a mapping did or did not apply.
Included / excluded	Element counts.	Quick completeness/audit signal.
Modified elements / relationships	Property-effect counts.	Shows non-existence transformations.
Applied variation effects	Number of rule applications.	Audit density and mapping execution.
Removed relationships	Relations whose endpoints/effects did not survive.	Traceability integrity audit.
Source revision	150% model revision used.	Current/stale determination.
Timestamp	Derivation time.	Evidence chronology.
Variant KPI snapshot	KPI values captured by derivation where available.	Early review; saved simulations remain the trade-study source.
Warnings / missing data	Assumptions, incomplete values or execution warnings.	Must be reviewed before treating evidence as decision-ready.

Derivation rule

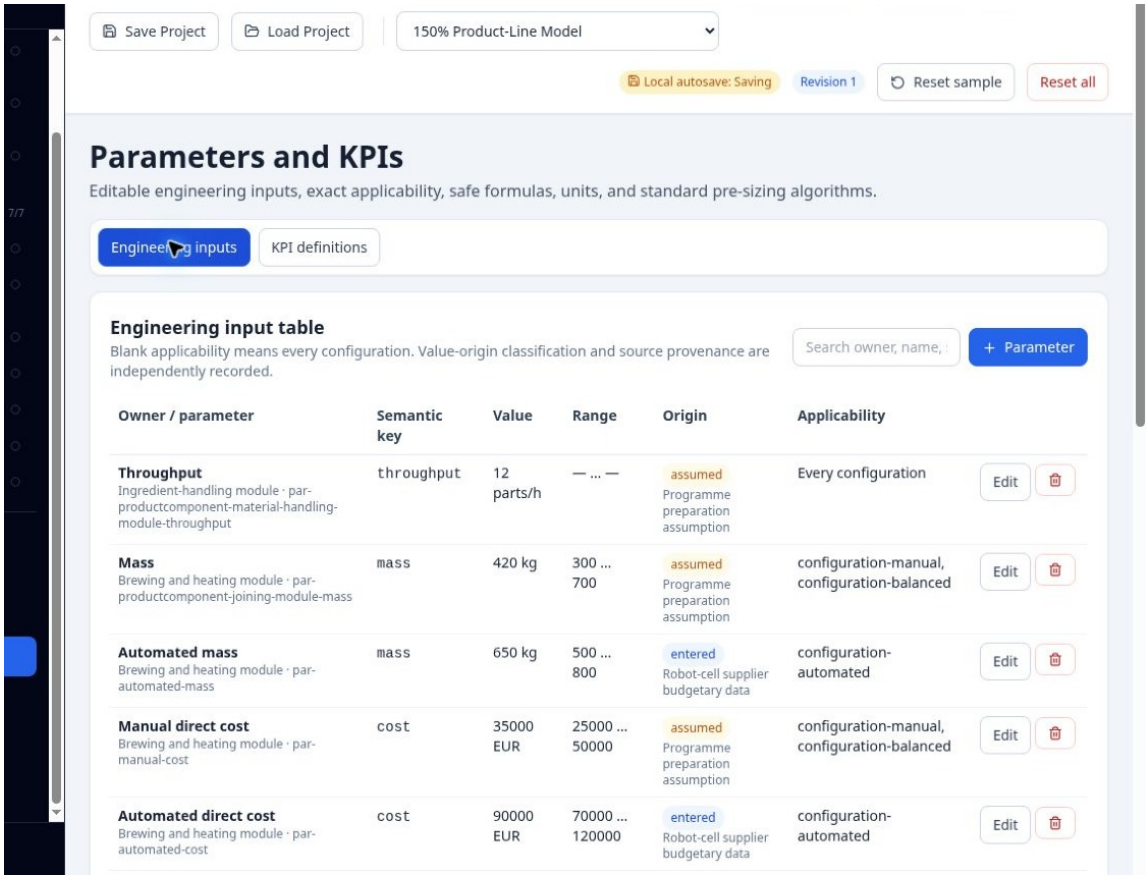
A saved 100% realization is fully independent evidence after derivation, but it is not an editable second source model in this page. Change the 150% source or configuration, then explicitly create a new derivation.

Part V - Parameters and KPIs

Engineering inputs are model-owned values. KPI definitions are reusable calculation recipes. Simulation combines a selected model/configuration with selected KPI definitions and freezes the exact inputs and results.

Engineering inputs tab

Field or control	What to enter or select	Allowed values / rule	Actual effect
Search owner, name, semantic key	Type any fragment.	Free text; blank shows all.	VIEW only.
+ Parameter	Click, type an exact Owner element ID, then type the parameter name.	Owner must exist. New parameter is number, null, entered, all configurations.	RUN/MODEL. This prompt-based route is best when you know the stable owner ID; the element side editor is friendlier.
Owner / parameter	Read owner name, parameter name and parameter ID.	Read-only.	READ. The parameter ID is used in KPI formulas.
Semantic key	Read the normalized engineering meaning.	Read-only in table; editable through Edit prompt.	RUN. Standard algorithms search exact semantic keys.
Value	Read nominal value and unit.	Read-only in table; editable through Edit prompt or element details.	READ/RUN.
Range	Read min, max and uncertainty.	Read-only summary.	READ.
Origin	Read entered, assumed, calculated or simulated plus provenance.	Read-only summary.	GOV/READ.
Applicability	Read Every configuration or exact configuration IDs.	Read-only summary.	READ/RUN.
Edit	Click and complete the sequence of browser prompts documented below.	Existing parameter.	RUN/GOV.
Delete	Click and confirm. The warning reports KPI formula references that may become invalid.	Existing parameter.	RUN/MODEL, destructive.



Engineering inputs centralize parameter identity, semantic keys, range, origin, applicability, process duration and resource quantities.

Parameter Edit prompt sequence

Prompt order	What to type	Validation and impact
1. Parameter name	Human-readable label.	Non-empty. Friendly display name.
2. Semantic key	Stable semantic classification, for example mass, directCost or power.	RUN. Case/spelling must match the standard algorithm's expected key.
3. Typed value	Number; blank means null.	Numeric parameters require a finite number or blank.
4. Unit	Compatible unit symbol.	RUN. Used by formulas and algorithms.
5. Recommended minimum	Finite lower bound or blank.	Must not exceed maximum.
6. Recommended maximum	Finite upper bound or blank.	Must not be below minimum.
7. Uncertainty percent	0-100 or blank.	Range evidence; nominal run uses the value.
8. Value origin	entered, assumed, calculated or simulated.	Exact lowercase choice required.
9. Source / provenance	Document, supplier, model or assumption source.	GOV. Blank allowed.
10. Applicable configuration IDs	Comma-separated exact IDs; blank means every configuration.	Unknown IDs are rejected. Use Element details checkboxes when possible.

Prompt cancellation

Pressing Cancel in a browser prompt stops that edit sequence. Previously typed prompt values are not partially committed by the function.

Authoritative process duration

Each card represents one Process function. Click it, type a positive duration, then type exactly minute, hour or day. The same metadata is editable in Model. One engineering day equals eight hours.

Industrial-component resource quantity

Each card represents Industrial-system component -> requiresResource -> Resource. Click it and enter a finite quantity greater than zero. Process cost and demand use the durations of process functions realized by that component.

Authoritative chain
Process function duration -> realizedBy -> industrial component -> requiresResource(quantity) -> resource(hourly rate). A missing link prevents a trustworthy process-cost/resource calculation.

KPI definitions tab

A KPI is not a stored result. It is a reusable definition containing objective links, calculation mode, formula or algorithm, output unit, optimization direction, weight and exact input references.

Create KPI prompt sequence

Prompt	Allowed content	Effect
KPI name	Required text.	Human-readable definition identity.
Mode	formula or standardAlgorithm.	Selects safe formula parser or locked algorithm.
Formula	For formula mode, for example param("parameter-id").	Parsed immediately; exact parameter/KPI IDs are collected.
Algorithm	For standardAlgorithm, one of the nine documented keys.	Locked calculation logic.
Output unit	Unit string or blank for dimensionless.	RUN. Used in result and formula validation.
Defaults	Direction minimize, weight 1, no objectives.	You must edit/link objectives before dashboard/trade eligibility.

KPI card controls

Field or control	What to enter or select	Allowed values / rule	Actual effect
Objectives measured	Check one or more objectives this KPI measures.	Any project objective.	MODEL/RUN. Required for dashboard and trade-study KPI eligibility.
Preview	Click to calculate this KPI in the active architecture-compatible model. KPI dependencies are included for preview.	Existing valid KPI.	READ/RUN preview only; no saved SimulationRun is created.
Edit metadata - formula KPI	Click to open formula editor.	Formula KPI only.	RUN. Current page does not expose formula KPI name/unit/direction/weight in this editor; use guided Trade Study setup for those fields.
Edit metadata - standard KPI	Complete prompts for name, description, output unit, direction and weight.	Direction minimize/maximize; finite weight.	RUN/GOV.
Delete	Click and confirm.	Deletion may be blocked when historical evidence references the KPI.	RUN/GOV, destructive when allowed.
ID	Read the stable KPI ID.	Read-only.	READ. Used by kpi("id") references.
Parameter IDs	Read collected parameter references.	Read-only.	READ.
Required KPI dependencies	Read referenced KPI IDs.	Read-only.	READ. All dependencies must also be selected in Simulation.
Standard algorithm explanation	Expand Logic, Inputs, Output and Assumptions.	Standard KPI only.	READ. This is the implemented formula contract.

Save Project Load Project 150% Product-Line Model

Local autosave: Saved Revision 1 Reset sample Reset all

Parameters and KPIs

Editable engineering inputs, exact applicability, safe formulas, units, and standard pre-sizing algorithms.

Engineering inputs KPI definitions

KPI definitions

Formula references are exact stable IDs; no display-name matching or dynamic JavaScript is used.

+ KPI

Formula editor · Handling Rate Formula

Search the canonical model and insert exact immutable reference tokens.

param("par-productcomponent-material-handling-module-throughput")

Search parameters and KPIs

Validate and save formula

Ingredient-handling module / Throughput param("par-productcomponent-material-handling-module-throughput")

Brewing and heating module / Mass param("par-productcomponent-joining-module-mass")

Total Mass standardAlgorithm algorithm locked

totalMass

Output kg · minimize · weight 1 · kpi-mass

OBJECTIVES MEASURED BY THIS KPI

Preview Edit metadata

The formula editor inserts exact parameter/KPI IDs. Display-name search is only a discovery aid; the stored token is immutable.

KPI formula editor - step by step

1	Open or create formula KPI In KPI definitions, click + KPI. Type Beverage mass total, type formula for Mode, enter 1 as the temporary formula, and type kg for Output unit. Why: The temporary literal lets the definition be created before references are inserted. Done when: The formula editor opens.
2	Find an input In Search parameters and KPIs, type mass. Find the intended component/parameter and click its token button. Why: Search uses friendly names; the inserted token stores the parameter ID. Done when: The expression contains param("...").
3	Add arithmetic Type + or use the guided operator buttons where available, then insert the next mass parameter. Why: Addition requires equal units. Done when: The formula has two exact references.
4	Validate and save Click Validate and save formula. Correct syntax, missing ID or unit errors reported by the safe parser. Why: Only validated syntax and collected references should be used in simulations. Done when: The editor closes and the card lists parameter IDs.
5	Link the objective Check Keep the appliance compact and efficient under Objectives measured by this KPI. Why: Trade-study eligibility requires at least one selected-scope objective link. Done when: The objective checkbox remains checked.
6	Preview, then simulate Click Preview for a quick current-context check. Use Simulation for immutable evidence. Why: Preview is not a saved run. Done when: Preview reports a value or a specific missing-input warning.

KPI formula grammar

Construct	Syntax	Rule
Parameter reference	param("parameter-id")	Exact stable ID; numeric active parameter must be available for the evaluated configuration.
KPI dependency	kpi("kpi-id")	Dependency KPI must be selected in Simulation and must not create a cycle.
Number	1; 2.5; -3	Dimensionless literal.
Arithmetic	+, -, *, /	Standard precedence; division by zero is an error.
Grouping	(expression)	Controls precedence.
Minimum / maximum	min(a,b); max(a,b)	At least one argument; all arguments must have equal units.
Sum / average	sum(a,b,c); average(a,b,c)	At least one argument; all arguments must have equal units.
Not supported	JavaScript, arbitrary function names, display-name lookup	Safe parser only; no eval.

How the formula parameter selector works

- The selector scans numeric parameters stored on model elements. Custom columns are not scanned.
- Search matches owner element, parameter name, parameter ID, unit, element type and tags in the guided builder.
- The friendly card is only for discovery. Clicking it inserts param("stable-id").
- When the guided builder is opened from Trade Study setup, it can show whether each parameter exists for the selected alternative configurations.
- KPI cards are also offered as kpi("stable-id") except the KPI currently being edited.
- Validation collects inputParameterIds and dependsOnKpids from the parsed formula. Do not manually maintain these lists.
- Renaming the owner, parameter or KPI does not break the token. Deleting the referenced object does.

Formula examples

Purpose	Formula	Expected output consideration
Add equal-unit masses	param("mass-a") + param("mass-b")	Both sources must use exactly compatible/equal units for this KPI parser.
Average cycle observations	average(param("cycle-a"), param("cycle-b"))	Both sources must have the same unit.
Cost plus process cost KPI	kpi("direct-cost") + kpi("process-cost")	Select all three KPIs in Simulation; units must match.
Scaled estimate	param("base-cost") * 1.15	Output retains base-cost dimension.
Throughput proxy from time	1 / param("cycle-hours")	Output dimension is inverse time; use an appropriate output unit.

Standard pre-sizing algorithms

Algorithm key	Implemented logic	Required model evidence	Output / key assumption
totalMass	Sum active numeric parameters whose semanticKey is mass.	Active mass parameters with compatible units.	Inherited unit; all included mass units compatible.
directElementCost	Sum active numeric parameters whose semanticKey is directCost.	Active direct-cost parameters.	Inherited currency unit.
processCost	For each industrial-component resource assignment: sum duration of realized process functions * hourlyRate * requiredQuantity.	Process duration; realizedBy; component requiresResource; resource hourlyRate.	EUR in current pre-sizing; day = 8 h; missing quantity defaults to 1 with warning.
estimatedTotalCost	directElementCost + processCost.	Both component calculations available.	EUR in current model.
totalPower	Sum active numeric parameters whose semanticKey is power.	Active power parameters with compatible units.	Inherited power unit.
manufacturingLeadTime	Longest path through active process-function precedes relations.	Positive durations and acyclic precedence sequence.	Hours; parallel branches use critical path; day = 8 h.
resourceDemand	Sum duration * requiredQuantity for component resource assignments; retain per-resource breakdown.	Duration, realizedBy, requiresResource quantity.	resource-hours; missing quantity defaults to 1 with warning.
basicUtilization	resource demand / available capacity.	Demand plus resource capacityHours and availabilityPercent in stored metadata.	% simplified planning utilization, not a detailed schedule.
throughputProxy	1 / manufacturingLeadTimeHours.	Valid lead-time result.	1/h comparative proxy, not actual production throughput.

Semantic-key spelling

The current standard algorithms look for mass, directCost and power. The sample includes some inherited early-demonstrator keys such as cost; inspect calculation warnings and use the exact required semantic key in new models.

Coffee sample KPI interpretation

Sample KPI	Mode	Direction	What to inspect
Total Mass	totalMass	minimize	Applicable mass parameters for Essential Capsule, Balanced Bean-to-Cup and Premium Dual Boiler.
Estimated total cost	estimatedTotalCost	minimize	Direct cost keys plus industrial process/resource cost chain.
Manufacturing Lead Time	manufacturingLeadTime	minimize	Coffee-machine manufacturing sequence and process durations.
Throughput Proxy	throughputProxy	maximize	Comparative reciprocal of lead time; not beverages/hour physical throughput.
Total Power	totalPower	minimize	Active power parameters by configuration.
Resource Demand	resourceDemand	minimize	Resource assignment quantities and process durations.
Basic Utilization	basicUtilization	minimize in sample	Demand versus stored capacity/availability assumptions.
Handling Rate Formula	formula	maximize	Exact reference to ingredient-handling Throughput parameter; inherited display unit may still say parts/h in the sample.

Part VI - Simulation

Simulation is deterministic pre-sizing over the selected model snapshot. It is not a dynamic physics or discrete-event solver. Every successful run stores an immutable input snapshot, selected definitions, results, warnings and revision.

Prepare run fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Analysis mode	Choose Configured 100% model or Architecture-only 150% model.	Two radio options.	RUN. Selects whether evidence comes from a configuration derivation/background realization or canonical unconfigured model.
Configuration	For configured mode, choose an active configuration.	Any active configuration, with validation status shown.	RUN. The system uses explicit current derivation where available and validates readiness.
Architecture context	For architecture-only mode, choose an active architecture.	Any non-archived architecture.	RUN. Evaluates the compatible 150% architecture context without a configuration.
Run name	Type a unique descriptive label before running.	Free text; non-empty recommended.	GOV/DOC. Identifies immutable history; does not change calculation.
KPI definitions	Check every KPI to calculate, including formula KPI dependencies.	Any stored KPI definitions.	RUN. Saved result rows are linked to KPI IDs.
Additional direct algorithms	Check an algorithm only when no saved KPI definition is required.	The same nine standard keys.	RUN. Result is algorithm-key evidence but not a reusable KPI definition.
Validate and run	Click after selecting model/context and at least one calculation.	Button action; validation can block execution.	RUN. On success creates immutable SimulationRun.

Prepare run selects configured 100% or architecture-only 150%, a run name, saved KPI definitions and optional direct algorithms.

Configured coffee run - smallest steps

1 Confirm realization currency

Open Variability > Configurator, select Essential Capsule and confirm Valid plus Current. If Stale, validate and derive again.

Why: Simulation evidence must match the current model revision.

Done when: Both badges are ready.

2 Open Simulation

Click Simulation in the left navigation.

Why: This opens run preparation and history.

Done when: Prepare run is visible.

3 Choose configured mode

Click Configured 100% model.

Why: Trade-study alternatives must be configuration-derived.

Done when: Configuration selector is visible.

4 Choose configuration

Select Essential Capsule.

Why: The run will use its effective features, property effects and applicable parameters.

Done when: The selector shows Essential Capsule.

5	Name the run Replace the default name with Essential Capsule - baseline pre-sizing. Why: A descriptive name makes history auditable. Done when: Run name field contains the label.
6	Select KPI definitions Check Total Mass, Estimated total cost, Manufacturing Lead Time and Throughput Proxy. If a formula KPI references another KPI, check that dependency too. Why: Trade study later requires the same selected KPI set for all alternatives. Done when: Four intended KPI boxes are checked.
7	Avoid duplicate direct algorithms Leave Additional direct algorithms unchecked when the same calculations already have saved KPI definitions. Why: Duplicate evidence complicates interpretation. Done when: No unnecessary direct algorithm is selected.
8	Run Click Validate and run. Read every error/warning. Repair model evidence rather than entering a fake fallback. Why: A run is only useful when its missing-information statements are understood. Done when: A new history item appears.
9	Repeat consistently Run Balanced Bean-to-Cup and Premium Dual Boiler with the same four KPI definitions. Why: Comparability requires the same calculation basis. Done when: Each configuration has a current complete run.

Read-only run history and result detail

Field or control	What to enter or select	Allowed values / rule	Actual effect
Run history item	Click a run name.	Newest and older runs; never editable.	VIEW. Selects immutable detail.
Timestamp / model revision	Read when and from which revision the run was made.	Read-only.	READ/GOV.
Current / Stale	Read status.	Current when source basis still matches; otherwise Stale.	READ. Stale evidence is retained but should not be used as current trade input.
Architecture / configuration	Read the exact analyzed context.	Read-only.	READ.
Variation-point count and realization type	Read explicit, background/in-memory or unconfigured context.	Read-only.	READ.
Result value / unit	Read numeric outcome or Not available.	Read-only.	READ. Not available is not zero.
Formula / algorithm	Read the executed definition.	Read-only.	READ/GOV provenance.
Input sources	Read each input name, value, unit and provenance.	Read-only.	READ. Main trace to engineering evidence.
Assumptions	Read implemented assumptions.	Read-only.	READ/GOV.
Missing information	Read absent inputs/links.	Read-only.	READ. Blocks trustworthy use even when another result exists.
Warnings	Read fallbacks, unit concerns or approximation limits.	Read-only.	READ/GOV.
Breakdown	Read component/resource contribution by element.	Shown when algorithm provides it.	READ. Supports diagnosis.
Critical chain	Read the process functions on longest path.	Shown for lead time.	READ. Explains lead-time result.

Save Project

Load Project

150% Product-Line Model

Local autosave: Saved

Revision 1

Reset sample

Reset all

input snapshot.

Premium Dual Boiler — comparable simulation

8/17/2026, 10:23:09 AM · rev 1 Current

Balanced Bean-to-Cup — comparable simulation

8/17/2026, 10:23:09 AM · rev 1 Current

Essential Capsule — comparable simulation

8/17/2026, 10:23:09 AM · rev 1 Current

3 variation point(s) captured · explicit realization.

Preliminary engineering estimate — not a verified detailed-design result. Source project revision 1.

Total Mass

KPI kpi-mass

650 kg

Formula / algorithm

totalMass

Input sources

Brewing and heating module / Automated mass: 650 kg (Robot-cell supplier budgetary data)

Assumptions

None

Missing information

None

Warnings

None

Estimated total cost

KPI kpi-cost

90016.45833333333 EUR

Formula / algorithm

estimatedTotalCost

Input sources

Brewing and heating module / Automated direct cost: 90000 EUR (Robot-cell supplier budgetary data); End-of-line test station → End-of-line test technician: 2 EUR/h (realized process duration 0.03333333333333333 h (Test completed coffee machine) × rate 60 × quantity 1); Coffee-machine integration cell → Automated module assembly cell: 12.083333333333332 EUR/h (realized process duration 0.08333333333333333 h (Integrate coffee-machine modules) × rate 145 × quantity 1); Module assembly workstation → Automated component feeder: 2.375 EUR/h (realized process duration 0.025 h (Manufacture coffee-machine modules) × rate 95 × quantity 1)

Assumptions

One engineering day is assumed to equal eight hours.

Missing information

None

Warnings

None

Manufacturing Lead Time

KPI kpi-lead-time

0.14166666666666666 h

A saved run retains values, exact formulas/algorithms, source inputs, assumptions, missing information, warnings, breakdown and critical chain.

Current versus stale

- Editing calculation-affecting model data increments the basis used to determine currency.
- Changing features, variation points, configurations, derivations, parameter values, formulas or KPI definitions can make later evidence stale.
- A stale run is preserved for history. It is not silently overwritten.
- Repair the source, validate/rederive as needed, then create a new run.
- Do not rename a stale run to imply currency; run status is computed from evidence links and revision.

Simulation troubleshooting order

Symptom	Check in this order	Repair
No configured run	Configuration active -> valid -> derivation current.	Validate and derive in Variability.
KPI Not available	Parameter applicability/value/unit -> formula reference -> algorithm semantic key.	Complete the exact missing input reported.
Formula dependency error	Every kpi("id") dependency selected; no cycle.	Select dependency or correct formula.
Process cost missing	Duration -> realizedBy -> requiresResource quantity -> hourly rate.	Complete canonical industrial chain.
Lead time wrong	Process duration units and sequence-specific precedes network.	Correct sequence and inspect critical chain.
Result stale	Model revision, derivation ID and KPI definition changes.	Rederive/re-run rather than editing history.

Part VII - Architecture Trade Study

The active Trade Study follows a strict order: mandatory requirement feasibility first, weighted KPI comparison second, explicit user decision last. It uses configuration-derived 100% alternatives with current complete saved simulations.

Top-level controls

Field or control	What to enter or select	Allowed values / rule	Actual effect
Manager view / Expert view	Choose the audience view before inspecting Step 2.	Two choices.	VIEW only. Manager shows outcome; Expert shows calculation/provenance.
New	Click to create a study and make it active.	Button action.	MODEL/RUN context.
Active Trade Study	Choose the study to edit/execute.	Any stored study.	MODEL/RUN context.
Study card	Click a named card.	Any stored study.	VIEW/MODEL context.
Duplicate	Copy the active study as an independent setup.	Active study.	MODEL. Results/history are governed by implementation copy rules; inspect before execution.
Delete	Click and confirm.	May be blocked by retained historical evidence.	MODEL/GOV, destructive when allowed.
1. Define study	Open framing, scope, axes and KPIs.	Active study.	VIEW.
2. Compare / Inspect evidence	Select eligible alternatives and inspect the current result.	Manager or Expert label depends on audience view.	VIEW/RUN selection context.
3. Decide	Create/approve explicit baseline decision.	Current result required.	GOV/RUN.
Run comparison	Validate readiness and create a new immutable result.	At least two selected eligible alternatives for competitive comparison.	RUN.
Saved immutable result	Choose a retained historical result.	Any result in active study.	VIEW. A stale result remains visible but cannot support a new approval.

Step 1 - Decision question fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Trade Study name	Type a concise study identity.	Non-empty text required for workflow completion.	DOC/MODEL. Stored name does not itself change numeric result and does not stale results.
Decision question	Type the explicit selection question.	Non-empty text; the placeholder Which valid architecture should be selected? is only a hint.	DOC/GOV framing. Stored question does not itself change scoring but is copied into decision evidence.

Problem-space scope fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Inherited system of interest	Read the designated stakeholder.	Exactly one required in Model.	READ/MODEL gate.
Project working scope	Read eligible use-case count.	Derived from Model working-scope selection and involvedIn.	READ/MODEL gate.
Use cases included	Check one or more working-scope use cases.	Only eligible use cases involving system of interest.	RUN/MODEL. Changes study scope and stales current result.
Eligible needs	Check needs in the selected use-case stakeholder context.	Filtered canonical needs.	RUN/MODEL. Changes mandatory scope.
Eligible objectives	Check objectives in the selected use-case stakeholder context.	Filtered canonical objectives.	RUN/MODEL. Controls KPI eligibility and mandatory scope.
Automatically traced requirements	Read requirements reached through derives from selected needs/objectives.	Read-only.	READ/RUN. These are mandatory feasibility requirements.

Scope recalculation

When model traceability changes, ineligible use cases, needs or objectives are removed and mandatory requirements are recomputed. This is intentional; repair canonical traces instead of re-entering out-of-scope IDs.

Variability axes and foundation fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Root Feature name	If no root exists, type a shared product-line root and click Create Root Feature.	Exactly one project root required.	MODEL/RUN.
Root name editor	When one root exists, edit its name directly.	Text.	MODEL/DOC. Stable feature ID remains.
Link shared Root	Click to link the active study to the project's one root.	One root only.	RUN/MODEL. Required for study readiness.
Variation axis name	Type the major design choice, for example Brewing architecture.	Required text.	MODEL/RUN. Creates a reusable axis and synchronized FeatureGroup.
Axis description	Explain what choice varies.	Free text.	DOC only.
Add axis	Click after root and name exist.	Button action.	MODEL/RUN.
Adopt existing group	Click Adopt when an unowned FeatureGroup should become an axis.	Eligible existing group.	MODEL/RUN.
Axis checkbox	Check axes explored by this study.	One or more required.	RUN. Changes study basis and stales results.
Axis name in card	Edit and leave the field to save.	Text.	MODEL/DOC. Synchronizes linked group name.
Axis description in card	Edit and leave the field to save.	Text.	DOC only.
Delete axis	Click and confirm.	Deletion may also remove an empty synchronized group.	MODEL/RUN.

Evaluation KPI fields in Trade Study setup

Field or control	What to enter or select	Allowed values / rule	Actual effect
KPI name	Type the criterion name.	Required text.	MODEL/DOC.
Calculation method	Choose Formula or Standard algorithm.	Two values.	RUN.
Standard algorithm	Choose one of nine algorithm keys.	Standard mode only.	RUN.
Output unit	Type expected unit.	Non-empty for valid trade KPI.	RUN.
Optimization direction	Choose Minimize or Maximize.	Two values.	RUN. Controls normalization direction.
Global weight	Enter a number at least zero.	At least one selected KPI must have positive weight.	RUN. Selected weights are normalized to percentages.
Objective measured	Choose a selected scoped objective.	Required for eligibility.	MODEL/RUN.
Guided Formula	Use exact tokens and validate.	Same safe KPI grammar.	RUN.
Create and select KPI	Click after valid fields/formula.	Button action.	MODEL/RUN. Adds reusable KPI and selects it in study.
Existing KPI select	Check an eligible KPI.	KPI must link to a selected study objective.	RUN. Includes it in simulation coverage and comparison.
Normalized weight note	Read percentage of total selected weight.	Read-only.	READ.
Objective checkboxes	Link/unlink scoped objectives.	Any selected-scope objectives.	RUN/MODEL eligibility.

The screenshot displays the TRACEABLE DECISION WORKFLOW interface for an Architecture Trade Study. The top navigation bar includes 'Save Project', 'Load Project', and a dropdown menu showing '150% Product-Line Model'. Below this, there are buttons for 'Local autosave: Saved', 'Revision 1', 'Reset sample', and 'Reset all'. The main workspace is titled 'Architecture Trade Study' and contains a sidebar on the left with a list of trade studies. The main area has tabs for '1. Define study', '2. Compare', and '3. Decide'. The 'Define study' tab is active, showing a 'COFFEE-MACHINE ARCHITECTURE SELECTION' study. The 'Decision question' section asks 'Which coffee-machine architecture should beco'. The 'Problem-space scope' section shows 'Coffee Machine System' as the inherited system of interest and '2 eligible use cases' as the project working scope.

Define study combines decision framing, trace-derived problem scope, one shared root, reusable axes and objective-linked KPI definitions.

Coffee study setup - smallest steps

1	Create/activate study Click New. In Trade Study name type Coffee-machine architecture selection. In Decision question type Which coffee-machine architecture should become the product-line baseline? Why: The question frames the final decision record. Done when: The study card displays the name.
2	Select use cases Under Use cases included, check the coffee-machine use cases relevant to the decision. Why: The study scope is a subset of the project working scope. Done when: Eligible needs/objectives are populated.
3	Select needs/objectives Check Fast beverage preparation and Consistent beverage quality, plus Serve peak household demand and Keep the appliance compact and efficient as appropriate. Why: Derived requirements and KPI eligibility depend on these choices. Done when: Automatically traced requirements lists coffee requirements.
4	Link root Confirm the one root is Coffee Machine Product Line. Click Link shared Root if not already linked. Why: Every axis and configuration uses the one shared root. Done when: Button reads Root linked.
5	Select axes Check existing Loading method, Assembly method and Inspection method axes, or create/adopt the coffee-equivalent groups used by your study. Why: Only selected axes define the planned trade space. Done when: Axis cards show selected styling.
6	Select KPI set Select Total Mass, Estimated total cost, Manufacturing Lead Time and Throughput Proxy. Confirm each measures a selected objective and has non-negative weight; keep at least one weight positive. Why: The same set must exist in every alternative run. Done when: No KPI eligibility or total-weight error remains.
7	Validate formulas/algorithms Expand each formula or standard algorithm. Confirm output units and directions: mass/cost/lead time minimize; throughput maximize. Why: Direction and unit affect normalized preference. Done when: Every selected KPI has a defined calculation and unit.

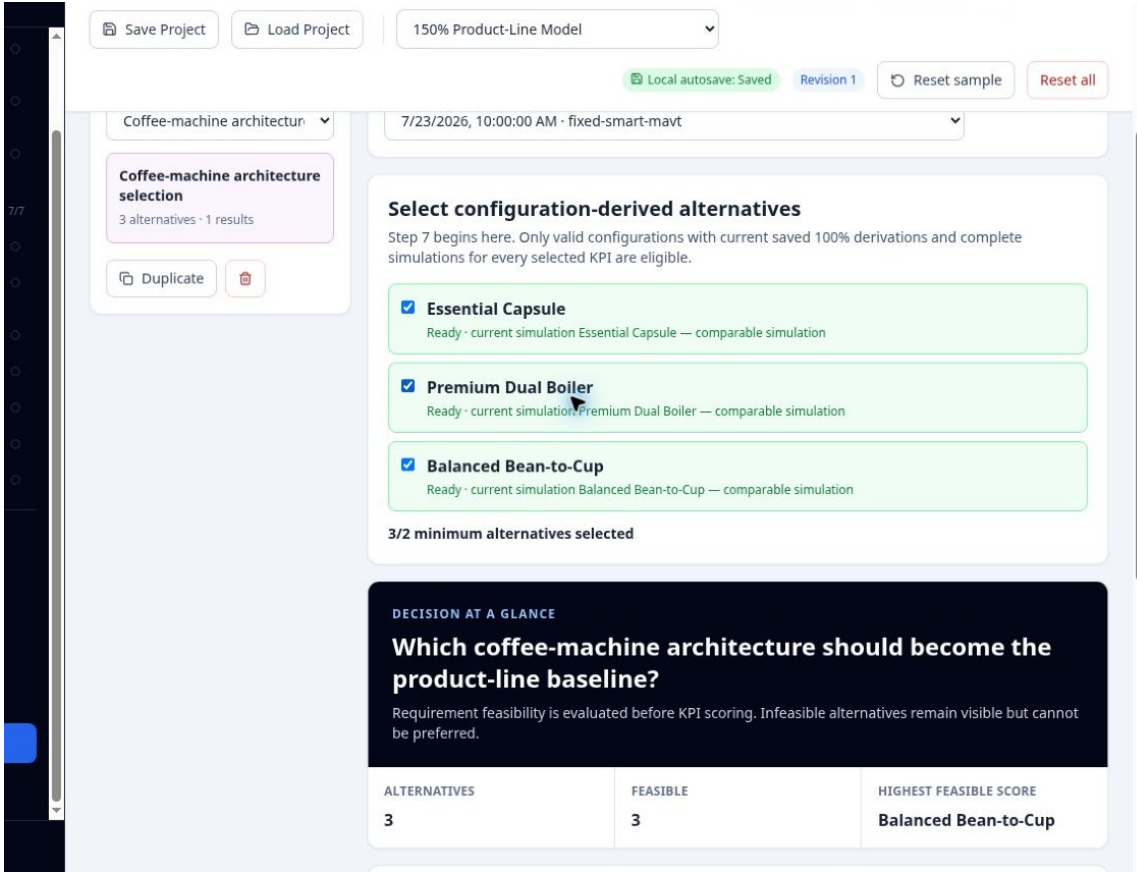
Step 2 - Select alternatives and compare

Field or control	What to enter or select	Allowed values / rule	Actual effect
Alternative checkbox	Check a configuration-derived alternative.	Enabled only with active valid configuration, current explicit 100% derivation and current complete run covering every selected KPI.	RUN. Adds candidate and alternative refs with simulation run ID.
Readiness message	Read Ready or Blocked explanation.	Read-only.	READ. Repair at source; no override exists here.
Use latest current run	Click if a selected alternative has a newer compatible run.	Selected ready alternative.	RUN. Updates exact evidence reference and states previous comparison basis.
Minimum count	Read selected count / 2.	At least two for competitive comparison.	READ/RUN gate.

Comparison methodology

Stage	Implemented behavior	Interpretation
1. Feasibility	Evaluate every mandatory requirement formula against each alternative's current evidence.	Failed/pending/error requirements make an alternative infeasible or unknown; it cannot be preferred as feasible.
2. Raw KPI	Read selected KPI values from the alternative's referenced SimulationRun.	The raw value is the engineering evidence.
3. Normalize	Convert each KPI to 0-100 using direction and observed alternative range.	Relative score is study-set dependent, not a physical unit.
4. Weight	Normalize selected non-negative global weights and multiply each normalized KPI score.	Weight expresses relative importance, not confidence.
5. Sum	Sum weighted contributions for each feasible alternative.	Weighted score is decision support after feasibility.
6. Preserve	Store immutable result with revision, settings timestamp, raw/normalized values and evidence.	Later changes mark it stale rather than rewriting history.

Manager view



Manager view shows the decision question, alternative count, feasible count, highest feasible score, feasibility/ranking and raw KPI comparison.

Read-only output	Meaning	Decision use
Alternatives	Number of selected configuration-derived alternatives.	Confirms comparison breadth.
Feasible	Alternatives satisfying mandatory requirement evidence.	Only these are decision-eligible.
Highest feasible score	Feasible leader(s) by weighted score.	Recommendation, not automatic decision.
Rank	Order among feasible alternatives.	Infeasible alternatives show no competitive rank.
Requirement feasibility	feasible/infeasible/unknown status.	Hard gate before preference.
Weighted score	0-100 relative aggregate.	Depends on current alternatives, directions and weights.
KPI comparison	Raw value/unit per selected KPI.	Check physical meaning before trusting score.

Expert view

Save Project

Load Project

150% Product-Line Model

Local autosave: Saving

Revision 1

Reset sample

Reset all

23T08:00:00.000Z

Alternative	KPI	Raw evidence	Normalized	Global weight	Contribution	Formula and provenance
Essential Capsule	Total Mass	420.00 kg	100.00	1	160.00	420 transform 80; raw weight 4e73-a281-60c90cfa8f4-66a4 totalMass.
Essential Capsule	Estimated total cost	35010.25 EUR	100.00	1	268.39	35010.25 transform 89; value function a3f8da54-edcc derivation-90c KPI formula est
Essential Capsule	Manufacturing Lead Time	0.18 h	0.00	1	200.00	0.1833333333 minimize value simulation-a3f6053efd5106b453c-85fa-44a manufacturing
Essential Capsule	Throughput Proxy	5.45 1/h	0.00	1	43.64	5.4545454545 value function simulation-a3f6053efd5106b453c-85fa-44a throughputPro
Balanced Bean-to-Cup	Total Mass	420.00 kg	100.00	1	160.00	420 transform 80; raw weight 47d5-bb67-228053a047-06e formula totalM
Balanced Bean-to-Cup	Estimated total cost	35016.46 EUR	99.99	1	268.37	35016.458333 value function simulation-4522ab379ccb9106eb-4bbb-9b estimatedTota
Balanced	Manufacturing	0.14 h	100.00	1	200.00	0.1416666666

Expert view exposes raw evidence, normalized value, weight, contribution, formula, run ID, derivation ID and requirement evidence.

Read-only output	Meaning	Review question
Immutable result ID / revision / time	Exact saved comparison evidence.	Is it current for the intended study basis?
Raw evidence	Saved simulation KPI value/unit.	Does it match the intended physical measure?
Normalized	0-100 direction-adjusted relative score.	Is the direction correct and is the observed range meaningful?
Global weight	Stored KPI importance before normalized share.	Does it reflect stakeholder priority?
Contribution	Normalized score times normalized weight.	Which KPI drives the aggregate?
Formula and provenance	Explanation plus run, derivation and KPI definition.	Can the result be traced to model inputs?
Requirement evidence	Requirement name, formula, status and actual evaluation message.	Is feasibility supported rather than inferred from a high score?

Step 3 - Decision and baseline

A comparison does not select a baseline. The user creates a decision, selects a feasible alternative, records ownership/rationale, confirms baseline effect and approves.

Field or control	What to enter or select	Allowed values / rule	Actual effect
Create decision from comparison	Click when a current result exists and no linked decision exists.	Button action.	GOV/RUN. Creates a draft linked to the immutable result/study.
Selected feasible alternative	Choose one explicit alternative.	Only alternatives with feasible requirement status.	GOV/RUN. Draft selection; changing it resets status to draft.
Owner	Type accountable person, team or board.	Free text.	GOV/DOC. Stored evidence; no numeric effect.
Rationale	Explain why this feasible alternative is selected, including trade-offs and limitations.	Required non-empty text for approval.	GOV. Decision evidence; no numeric effect.
Confirm baseline approval	Check that approval establishes the selected architecture baseline.	Required boolean.	GOV/RUN. Enables approval and baseline link.
Confirm selection	Click while draft selection exists.	Draft only.	GOV. Moves selection to proposed status.
Approve decision	Click after alternative, rationale and baseline confirmation are complete.	Current result required.	GOV/RUN. Sets approved status/date and selected architecture baseline.
Decision status	Read draft, proposed or approved.	Read-only badge.	READ/GOV.

Save Project Load Project 150% Product-Line Model

Local autosave: Saved Revision 1 Reset sample Reset all

TRACEABLE DECISION WORKFLOW

Manager view Expert view Preliminary engineering estimate — not a verified detailed-design result.

Architecture Trade Study

Requirement feasibility first; weighted KPI comparison second; explicit user decision last.

Trade Studies + New

ACTIVE TRADE STUDY

Coffee-machine architecture

Coffee-machine architecture selection

3 alternatives · 1 results

Duplicate

1. Define study 2. Inspect evidence 3. Decide Run comparison

SAVED IMMUTABLE RESULT

7/23/2026, 10:00:00 AM · fixed-smart-mavt

Decision and rationale

approved

Only requirement-feasible alternatives are eligible. One feasible alternative is sufficient, with an explicit no-competitive-ranking notice.

SELECTED FEASIBLE ALTERNATIVE

Balanced Bean-to-Cup

OWNER

Sample Product Governance Board

RATIONALE

Sample governance decision: mandatory feasibility, Pareto position, fixed stakeholder value, sensitivity and bounded robustness were reviewed together.

Decision approval is explicit. A rank never establishes a baseline without feasible selection, rationale and confirmation.

1 Use a current result

Choose the latest intended Saved immutable result and confirm no stale banner is shown.

Why: A stale result cannot support a new approval.

Done when: Current result is displayed.

2 Create draft

Open 3. Decide and click Create decision from comparison if needed.

Why: The decision snapshots the comparison context.

Done when: Decision fields appear with draft status.

3 Select feasible coffee alternative

Choose one feasible alternative, for example Balanced Bean-to-Cup, from Selected feasible alternative.

Why: Only feasible alternatives are offered.

Done when: The field displays the intended selection.

4 Record owner

Type Product Governance Board in Owner.

Why: Accountability is part of audit evidence.

Done when: Owner is stored.

5 Record rationale

Type a complete explanation: mandatory requirements passed; mass/cost/lead-time/throughput trade; limitations; actions before detailed design.

Why: The rationale explains why the score is acceptable, not merely which score is highest.

Done when: Rationale is non-empty.

6 Confirm baseline effect

Check Confirm that approval establishes the selected architecture baseline.

Why: Prevents accidental baseline creation.

Done when: Checkbox is selected.

7 Confirm and approve

Click Confirm selection when shown, review status, then click Approve decision.

Why: Approval is an explicit governance act.

Done when: Status is approved and baseline is linked.

One feasible alternative

The page permits an explicit decision when only one alternative is feasible, but displays a notice that no competitive feasible ranking exists. Document why the trade space collapsed.

Part VIII - Snapshots and Delivery

Export contains five tabs: Save / Load Project, Snapshots, Selective Export, PDF Report and Demo Checklist. Complete project JSON is the reloadable project artifact. Selective exports and PDF are review/delivery outputs.

Save / Load Project

Action	What happens	Use it when	Impact
Save Project	Downloads complete active project JSON after preparing the portable package.	Milestones, transfer and recovery.	GOV/persistence. Includes model and analysis evidence; excludes application snapshots/UI preferences.
Saved project file	Choose a .json or .mbse-project.json file.	Local file input.	VIEW until validation/load.
Load as new project	Keep current project and resolve only top-level project-ID collision.	Radio choice.	MODEL/persistence.
Replace active project	Preserve active project ID and create a safety snapshot first.	Radio choice with confirmation.	MODEL/persistence.
Validate and load	Validates package before mutation.	Button enabled after a file is read.	MODEL. Invalid input changes nothing.
Download raw input	Downloads the selected file text unchanged.	Shown after file is read.	GOV/diagnostic only.

Snapshots

Field or control	What to enter or select	Allowed values / rule	Actual effect
Required name	Type milestone name.	Non-empty; button disabled when blank.	GOV. Snapshot identity.
Optional note	Describe purpose, assumptions or milestone.	Free text.	DOC/GOV.
Create snapshot	Click to store application-level local snapshot.	Maximum 20 per source project.	GOV/persistence. Not enterprise version control.
Restore	Click and confirm. A safety snapshot is created first; at limit, replacement of oldest is confirmed.	Existing snapshot.	MODEL/persistence.
Duplicate as project	Click to create an independent project from snapshot data.	Existing snapshot.	MODEL/persistence.
Raw	Download snapshot JSON.	Existing snapshot.	GOV/delivery.
Delete snapshot	Click and confirm.	Existing snapshot.	Persistence, destructive.

Selective Export fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Scope	Choose Complete project or Selection.	Two values.	GOV/delivery scope.
Model domain	Choose All, Mission and requirements, Product, Process or Resources.	Five values.	GOV/delivery filter.
Architecture	Choose All or one architecture.	Stored architectures.	GOV/delivery filter.
Configuration	Choose All or one configuration.	Stored configurations.	GOV/delivery filter.
Element types	For Selection, check intended element types.	Any element types.	GOV/delivery filter.
Individual elements	Use multi-select to include exact elements.	Any model elements.	GOV/delivery filter.
Features	Use multi-select for exact features.	Any features.	GOV/delivery filter.
Relationship types	Check relation types.	Any canonical relation types.	GOV/delivery filter.
Tags	Check tags found in current elements.	Any current tag.	GOV/delivery filter.
Validation results	Check to include.	Boolean.	GOV/delivery.
KPI definitions	Check to include.	Boolean.	GOV/delivery.
Simulation results	Check to include.	Boolean.	GOV/delivery.
Comparisons	Check to include.	Boolean.	GOV/delivery.
Decisions	Check to include.	Boolean.	GOV/delivery.
Automatically include relationship endpoints	Choose to preserve selected relationship integrity by adding missing endpoints.	Radio option.	GOV. Preferred for trace-complete export.
Omit incomplete relationships and record them	Choose to exclude relations whose endpoints are outside selection and list omissions.	Radio option.	GOV. Transparent but intentionally incomplete scope.
Export preview	Read calculated counts, omissions and warnings before download.	Read-only.	READ/GOV.
JSON	Download selected package.	Preview must be valid.	GOV/delivery.
XLSX	Generate and download workbook.	Preview must be valid.	GOV/delivery. Review artifact, not reloadable project.

PDF Report fields

Field or control	What to enter or select	Allowed values / rule	Actual effect
Report title	Type report heading.	Free text; blank uses implementation default.	DOC/delivery.
Scope summary	Type concise report boundary.	Free text.	DOC/delivery.
Architecture	Choose one or Not selected.	Stored architectures.	Delivery selection.
Configuration	Choose one or Not selected.	Stored configurations.	Delivery selection.
Trade Study	Choose one or Not selected.	Stored studies.	Delivery selection.
Trade Study result	Choose one immutable result or Not selected.	Results in selected study.	Delivery selection.
Formal decision	Choose one or Not selected.	Stored decisions.	Delivery selection.
Validation	Check to include validation section.	Boolean.	Delivery selection.
Simulation detail	Check to include detailed run evidence.	Boolean.	Delivery selection.
Risks	Check to include risk content.	Boolean.	Delivery selection; current simplified Trade Study does not author all legacy risk fields.
Generate PDF	Click to download report.	Button action.	GOV/delivery. Missing selected content is reported honestly as unavailable.

Demo Checklist

The 14 checkboxes are a temporary in-page companion for demonstration flow. Checking them does not alter model, derivation, simulation, comparison or decision completeness. They are VIEW-only reminders and are not a substitute for dashboard status.

Recommended delivery routine

1	Run validation Open Model quality and click Run validation. Resolve or document all critical findings. Why: Delivery should not hide known quality state. Done when: Current findings are understood.
2	Create named snapshot Open Export > Snapshots, type a milestone name and note, then click Create snapshot. Why: Provides quick local rollback. Done when: Snapshot appears at top of list.
3	Save complete project Open Save / Load Project and click Save Project. Why: This is the reloadable portable artifact. Done when: JSON download completes.

4 Prepare review package

Open Selective Export, choose scope and optional records, preserve relationship endpoints, inspect preview, then download JSON and/or XLSX.

Why: Reviewers may need a smaller trace-complete package.

Done when: Manifest counts match intended scope.

5 Generate report

Open PDF Report, select architecture/configuration/study/result/decision, choose included sections and click Generate PDF.

Why: Provides readable evidence summary.

Done when: Report downloads and states unavailable content honestly.

6 Test reload

In a different project context, use Load as new project with the saved complete JSON and verify it opens.

Why: A backup is only useful when it can be restored.

Done when: Loaded copy passes package validation.

Part IX - Coffee-machine example reference

The loaded sample is a worked product-line study. Use it to learn navigation before creating a clean project. Its values are preliminary demonstration inputs, not a validated coffee-machine design.

Problem-space content

Type	Sample content	Canonical trace
Mission	Deliver a configurable coffee experience	hasStakeholder to the four context stakeholders.
System of interest	Coffee Machine System	Stakeholder with system-of-interest checkbox selected; involvedIn use cases.
Other stakeholders	Product Manager; Coffee Consumer; Service and Quality Manager	hasNeed, hasObjective and involvedIn as applicable.
Needs	Fast beverage preparation; Consistent beverage quality; Safe and intuitive operation	Stakeholder -> hasNeed -> Need -> derives -> Requirement.
Objectives	Serve peak household demand; Keep the appliance compact and efficient	Stakeholder -> hasObjective -> Objective -> derives -> Requirement; KPIs link to objectives.
Use cases	Configure coffee-machine variant; Manufacture coffee machine; Verify manufactured coffee machine	Stakeholder -> involvedIn -> Use case -> hasFunction -> product/process functions.
Requirements	Meet required beverage throughput; verify beverage quality; remain below mass limit	Derived from needs/objectives; satisfiedBy architecture evidence; numeric formula binding.

Product and industrial architecture

Domain	Function	Realizing technical component
Product	Meter coffee ingredients	Ingredient-handling module
Product	Heat and pressurize brew water	Brewing and heating module
Product	Monitor beverage quality	Beverage sensing module
Industrial system	Manufacture coffee-machine modules	Module assembly workstation
Industrial system	Integrate coffee-machine modules	Coffee-machine integration cell
Industrial system	Test completed coffee machine	End-of-line test station

Product item flow through manufacturing

Process function	Direction	Product component	Quantity / unit / name
Manufacture coffee-machine modules	consumes	Purchased component kit	1 kit - Purchased parts released to production
Manufacture coffee-machine modules	produces	Ingredient-handling module	1 module - Manufactured ingredient-handling module
Manufacture coffee-machine modules	produces	Brewing and heating module	1 module - Manufactured brewing and heating module
Manufacture coffee-machine modules	produces	Beverage sensing module	1 module - Manufactured beverage sensing module
Integrate coffee-machine modules	consumes	Three manufactured modules	1 module each - for integration
Integrate coffee-machine modules	produces	Integrated coffee machine	1 machine
Test completed coffee machine	consumes	Integrated coffee machine	1 machine - for end-of-line test
Test completed coffee machine	produces	Verified coffee machine	1 machine

Representative sample inputs

Owner	Input	Sample value	Effect
Ingredient-handling module	Throughput	12 parts/h in the inherited sample input	Requirement and Handling Rate Formula evidence.
Brewing and heating module	Mass	420 kg for Essential/Balanced; 650 kg for Premium	totalMass and mass requirement.
Brewing and heating module	Direct cost	35,000 EUR versus 90,000 EUR inputs	Cost evidence; confirm semanticKey expected by algorithm.
Brewing and heating module	Power	12 kW versus 38 kW	totalPower.
Beverage sensing module	Quality confidence	99%	Quality requirement.
Manufacture modules	Duration	4 min; variation rule may set 1.5 min for automated feed	Lead time, process cost and demand.
Integrate modules	Duration	5 min	Lead time, process cost and demand.
Test completed machine	Duration	2 min	Lead time, process cost and demand.
Manufacturing technician	Hourly rate	55 EUR/h	processCost through component requiresResource.
End-of-line test technician	Hourly rate	60 EUR/h	processCost through component requiresResource.
Automated module assembly cell	Hourly rate	145 EUR/h	Automated processCost.
Automated component feeder	Hourly rate	95 EUR/h	Automated processCost.

Sample naming note

Some values and units intentionally retain the earlier industrial-demonstrator basis, for example parts/h and large equipment-style mass/cost. Treat them as traceability mechanics, not realistic appliance sizing.

Requirement formulas in the sample

Requirement	Expression	Bound evidence
The machine shall meet the required beverage throughput	@throughput >= 10	Ingredient-handling module Throughput parameter.
The machine shall verify beverage quality	@quality_confidence >= 95	Beverage sensing module Quality confidence parameter.
The machine shall remain below the mass limit	@equipment_mass <= 1000	Brewing and heating module configuration-applicable mass parameter.

Coffee feature model and configurations

Feature/group	Type	Sample role
Coffee Machine Product Line	root	One shared family root.
Common Brew Control; Food and Electrical Safety	mandatory	Automatically effective common capabilities.
Capsule Input / Automatic Bean Grinder	xor	Loading/ingredient choice.
Single Thermoblock / Dual Boiler	xor	Brewing architecture choice.
Beverage Sensing / Milk Temperature Probe	or	Selectable inspection/sensing technologies.
Brew History; Adaptive Brew Profile	optional	Data and analytics choices.
Regional Voltage	mandatory enumeration	Allowed values 230V, 400V.

Configuration	Manual feature selections	Typed value
Essential Capsule	Capsule Input, Single Thermoblock, Beverage Sensing, Brew History	Regional Voltage = 230V
Balanced Bean-to-Cup	Automatic Bean Grinder, Dual Boiler, Beverage Sensing, Brew History	Copied sample value, inspect before use
Premium Dual Boiler	Automatic Bean Grinder, Dual Boiler, Beverage Sensing, Milk Temperature Probe, Brew History, Adaptive Brew Profile	Regional Voltage = 400V

Sample constraints and variation effects

Rule	Meaning	Observed effect
Automatic Bean Grinder requires Dual Boiler	Automated feed cannot coexist with the single-heater choice in the sample.	Configurator adds dependency or reports invalid selection.
Dual Boiler requires Food and Electrical Safety	Premium heating requires mandatory safety content.	Automatic/mandatory selection.
Adaptive Brew Profile requires Brew History	Analytics requires data recording.	Brew History becomes automatic if needed.
Capsule Input excludes Automatic Bean Grinder	Mutually exclusive ingredient input.	Configuration invalid if both effective.
Single Thermoblock excludes Dual Boiler	Mutually exclusive heating architecture.	Configuration invalid if both effective.
Manufacture duration by automation	Property variation on metadata:duration.	4 min for capsule/manual input, 1.5 min for automatic feed.
Test-data interface voltage	Typed enumeration changes description.	230 V or 400 V description in derived model.
Feature-controlled resources/components	Existence variation points include/exclude targets.	Variant-specific 100% architecture.

Part X - Build the coffee-machine study from scratch

This chapter deliberately breaks the work into the smallest practical actions. Enter the values exactly as shown for a tutorial project, then replace them with your domain evidence in a real project.

A. Create the project and mission context

A1	Create project Click the project selector area, click New project, type Coffee Machine Tutorial, and press OK. Why: Creates an independent source model. Done when: Active project shows Coffee Machine Tutorial.
A2	Choose scope Click Dashboard. Select Trade study so the full page storyline is visible. Why: The tutorial will exercise every workspace. Done when: Dashboard shows trade-study Problem and Solution Space.
A3	Open Mission table Click Model. Click Mission and Context. Click Editable table. Why: This is where the intent chain begins. Done when: Mission/Stakeholder/Need/Objective/Use case types are available.
A4	Create mission row In New element type choose Mission. Click Add row. In the side panel replace the name with Deliver a configurable coffee experience. Type a description. Select reviewed. Type owner Product Planning and source Tutorial. Click Save element. Why: Name and source make the mission auditable. Done when: The mission row persists.
A5	Create system stakeholder Choose Stakeholder and click Add row. Rename it Coffee Machine System. Check Designate this stakeholder as the system of interest. Set status reviewed. Click Save element. Why: Exactly one system of interest is required. Done when: System checkbox remains selected after reopen.
A6	Create other stakeholders Create Product Manager, Coffee Consumer, and Service and Quality Manager as Stakeholder rows. Save each. Why: Needs, objectives and use cases must be connected to interested parties. Done when: Four stakeholders exist.

A7	Link mission to stakeholders Scroll to Relationship editor. Select Deliver a configurable coffee experience as Source, hasStakeholder as Relationship, each stakeholder as Target, and click Create. Repeat four times. Why: Mission intent must reach the context actors. Done when: Mission has four connections.
A8	Create needs Choose Need and add Fast beverage preparation, Consistent beverage quality, and Safe and intuitive operation. Save each. Why: Needs capture stakeholder-valued outcomes. Done when: Three needs exist.
A9	Link needs Create Product Manager -> hasNeed -> Fast beverage preparation; Service and Quality Manager -> hasNeed -> Consistent beverage quality; Coffee Consumer -> hasNeed -> Safe and intuitive operation. Why: Trace defines who values each outcome. Done when: Each need has an incoming hasNeed.
A10	Create objectives Choose Objective and add Serve peak household demand plus Keep the appliance compact and efficient. Save each. Why: KPIs later measure objectives. Done when: Two objectives exist.
A11	Link objectives Create Coffee Machine System -> hasObjective -> both objectives. Why: Connects measurable targets to system intent. Done when: Both objectives are traced.

B. Define use cases and requirements

B1	Create use cases Choose Use case and create Configure coffee-machine variant, Manufacture coffee machine, and Verify manufactured coffee machine. Why: Use cases state observable services/interactions. Done when: Three use cases exist.
B2	Link system to use cases Create Coffee Machine System -> involvedIn -> each use case. Add other stakeholder involvedIn links where meaningful. Why: Only use cases involving the system can be selected. Done when: Each use case is eligible.

B3	Select working scope In the editable table, check In working scope for all three tutorial use cases. Why: Semantic scope drives dashboard and trade study. Done when: Three use-case checkboxes are selected.
B4	Open requirements Click Requirements and Validation. Choose System requirement. Why: Requirements are separated from intent but remain traced. Done when: System requirement is available as new row type.
B5	Create throughput requirement Click Add row. Rename to The machine shall meet the required beverage throughput. Add threshold/unit meaning in Description. Click Save element. Why: A requirement must be testable. Done when: Requirement exists.
B6	Create quality and mass requirements Create The machine shall verify beverage quality and The machine shall remain below the mass limit. Save both. Why: These become mandatory feasibility evidence. Done when: Three requirements exist.
B7	Derive requirements In Relationship editor create Fast beverage preparation -> derives -> throughput requirement; Serve peak household demand -> derives -> throughput requirement; Consistent beverage quality -> derives -> quality requirement; Safe and intuitive operation -> derives -> mass requirement; Keep the appliance compact and efficient -> derives -> mass requirement. Why: Trade-study mandatory requirements are collected through derives. Done when: Every requirement has at least one derives source.
B8	Create verification methods Choose Verification method and create Manufacturing cycle-time analysis and End-of-line functional test. Save each. Why: Verification method records how evidence will be obtained. Done when: Two methods exist.
B9	Link verification Create Manufacturing cycle-time analysis -> verifies -> throughput requirement and End-of-line functional test -> verifies -> quality requirement. Why: Verification trace is governance evidence. Done when: Verification links exist.

C. Build product functional and technical architecture

C1	Create product functions Open Product Functional. Create Meter coffee ingredients, Heat and pressurize brew water, and Monitor beverage quality. Why: Functions describe behavior before structure. Done when: Three product functions exist.
C2	Connect use cases Create Manufacture coffee machine -> hasFunction -> all three product functions. Add other use-case links if their behavior uses a function. Why: Dashboard product-functional evidence is use-case based. Done when: Every selected use case intended for product behavior has a function link.
C3	Create product components Open Product Technical. Create Ingredient-handling module, Brewing and heating module, Beverage sensing module, Purchased component kit, Integrated coffee machine and Verified coffee machine. Why: These are the product technical architecture and manufacturing item-flow objects. Done when: Six product components exist.
C4	Realize product functions Create Meter coffee ingredients -> realizedBy -> Ingredient-handling module; Heat and pressurize brew water -> realizedBy -> Brewing and heating module; Monitor beverage quality -> realizedBy -> Beverage sensing module. Why: Every function must be allocated to structure. Done when: Three realizedBy links exist.
C5	Trace requirements to architecture Create each requirement -> satisfiedBy -> its relevant product function and/or product component. Why: Allows traceability recap and parameter-binding helper to reach architecture evidence. Done when: Requirement cells show satisfying content.
C6	Create product interfaces Choose Product interface and add Ingredient and water interface plus Thermal and hydraulic interface. Connect relevant functions/components with connects. Why: Interfaces document boundaries; they do not calculate by themselves. Done when: Interface connections are visible in the graph.

D. Build industrial-system functional and technical architecture

D1	Create process functions Open Process Functional. Create Manufacture coffee-machine modules, Integrate coffee-machine modules and Test completed coffee machine. Why: These describe the industrial behavior. Done when: Three process functions exist.
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D2	Enter durations For Manufacture type 4 minute; Integrate type 5 minute; Test type 2 minute in Authoritative duration. Why: Time-based algorithms need explicit units. Done when: All three durations are visible.
D3	Connect use cases Create Manufacture coffee machine -> hasFunction -> Manufacture modules and Integrate modules; Verify manufactured coffee machine -> hasFunction -> Test completed machine. Why: Dashboard industrial-functional evidence is use-case based. Done when: Relevant use cases have process functions.
D4	Create industrial components Open Process Technical. Create Module assembly workstation, Coffee-machine integration cell and End-of-line test station. Why: These are the technical means that realize process behavior. Done when: Three industrial components exist.
D5	Realize process functions Create Manufacture modules -> realizedBy -> Module assembly workstation; Integrate modules -> realizedBy -> Coffee-machine integration cell; Test machine -> realizedBy -> End-of-line test station. Why: Algorithms follow this realization chain. Done when: Every process function is realized.
D6	Create process interfaces Create Component logistics interface and Test-data interface. Connect appropriate process functions/industrial components with connects. Why: Documents industrial connectivity. Done when: Interfaces appear in process graph.
D7	Create resources Create Manufacturing technician, End-of-line test technician, Automated module assembly cell and Automated component feeder as Resource elements. Set Resource type and Hourly rate 55, 60, 145 and 95. Why: Resources and rates feed process cost/demand. Done when: Four resources exist with rates.
D8	Allocate resources correctly Create Module assembly workstation -> requiresResource -> Manufacturing technician; integration cell -> requiresResource -> Manufacturing technician; test station -> requiresResource -> End-of-line test technician. Enter quantity 1 and unit person. Add automated resource links from the relevant industrial components with quantity 1 and unit machine. Why: Resources belong to realizing components, never directly to functions. Done when: requiresResource source is always an industrial component.

D9 Create process sequence

Open Process Functional > Function sequence. Click Create sequence. Type Coffee-machine manufacturing process. Select Manufacture coffee machine. Check all three leaf functions. Link Manufacture -> Integrate and Integrate -> Test.

Why: Lead time uses sequence-specific precedes.

Done when: Sequence valid with one start and one end.

D10 Create item flows

In Product input/output flows add Purchased kit consumed by Manufacture; three modules produced by Manufacture and consumed by Integrate; Integrated machine produced by Integrate and consumed by Test; Verified machine produced by Test. Use positive quantity 1 and units kit/module/machine.

Why: Handoff validation connects product and industrial architectures.

Done when: No missing product handoff message remains.

E. Add parameters and executable requirements**E1 Add throughput**

Open Ingredient-handling module details. Click Add under Parameters. Rename to Throughput. Enter 12, unit parts/h, origin assumed and source Tutorial preliminary input.

Why: Requirement and formula KPI need numeric evidence.

Done when: Throughput card is complete.

E2 Add mass

Open Brewing and heating module. Add Mass = 420 kg, origin assumed, source Tutorial preliminary input, minimum 300 and maximum 700.

Why: Mass feeds requirement and totalMass.

Done when: Mass card is complete.

E3 Add quality

Open Beverage sensing module. Add Quality confidence = 99 %, origin assumed and a source.

Why: Quality requirement needs evidence.

Done when: Quality parameter is complete.

E4 Bind throughput requirement

Open throughput requirement. Select Ingredient-handling module - Throughput. Click Insert. Complete expression @throughput >= 10. Click Save element.

Why: Binding stores exact ID.

Done when: Status is satisfied.

E5 Bind quality requirement

Insert Quality confidence and complete @quality_confidence >= 95. Save.

Why: Creates executable quality gate.

Done when: Status is satisfied.

E6	Bind mass requirement Insert Mass and complete @equipment_mass <= 1000. Save. Why: Creates executable feasibility gate. Done when: Status is satisfied.
E7	Run validation Open Model quality and click Run validation. Fix identity, relationship, traceability, process, parameter and formula errors. Why: Later analyses depend on complete canonical evidence. Done when: No unexplained error remains.

F. Define variability and 100% alternatives

F1	Create root Open Variability > Feature Model. Click Feature. Type Coffee Machine Product Line and choose root. Save. Why: Exactly one root is required. Done when: Root appears.
F2	Create mandatory features Create Common Brew Control and Food and Electrical Safety as mandatory children of the root. Why: Common behavior is automatic. Done when: Two mandatory children exist.
F3	Create loading XOR Create Capsule Input and Automatic Bean Grinder as xor children with the same Cardinality group ID loading-method. Why: Exactly one loading approach is effective. Done when: Two XOR members exist.
F4	Create brewing XOR Create Single Thermoblock and Dual Boiler as xor children with group ID brewing-method. Why: Exactly one heating architecture is effective. Done when: Two XOR members exist.
F5	Create sensing OR Create Beverage Sensing and Milk Temperature Probe as or children with group ID sensing-options. Why: One or more sensing choices may be effective. Done when: Two OR members exist.

F6	Create optional data features Create Brew History and Adaptive Brew Profile as optional children. Why: Independent selectable capabilities. Done when: Two optional features exist.
F7	Create voltage enumeration Create Regional Voltage as mandatory enumeration. Type 230V, 400V in Allowed values and select 230V as Default. Why: Typed feature values can drive property effects. Done when: Enumeration has two values.
F8	Add constraints Create Automatic Bean Grinder requires Dual Boiler; Dual Boiler requires Food and Electrical Safety; Adaptive Brew Profile requires Brew History; Capsule Input excludes Automatic Bean Grinder; Single Thermoblock excludes Dual Boiler. Why: Cross-tree rules make selections valid. Done when: Five constraints exist.
F9	Create existence mappings Open Variation Points. For each variant-specific resource/component, create Existence mapping, choose target, insert its feature ID in ImpactedByFeature, choose appropriate domain and leave Enabled checked. Why: Feature selection must create a physical architecture effect. Done when: Mappings appear in list.
F10	Create duration mapping Create Primitive Property mapping Loading duration by automation. Target Manufacture coffee-machine modules. Safe property path metadata:duration. Add rule Automatic Bean Grinder -> 1.5 and rule Capsule Input -> 4. Set domain process. Why: Demonstrates parameterized physical/process effect. Done when: Two ordered rules are visible.
F11	Create voltage property mapping Create Primitive Property mapping on Test-data interface description. Add typed Regional Voltage equals 230V and 400V rules with explanatory text. Why: Demonstrates enumeration-driven 100% property. Done when: Two typed rules exist.
F12	Create Essential Open Configurator. New configuration Essential Capsule. Select Capsule Input, Single Thermoblock, Beverage Sensing and Brew History; choose 230V. Validate and derive. Why: First alternative. Done when: Valid and Current.

F13	Create Balanced Duplicate Essential. Rename Balanced Bean-to-Cup. Select Automatic Bean Grinder, Dual Boiler, Beverage Sensing and Brew History; choose intended voltage. Validate and derive. Why: Second alternative. Done when: Valid and Current.
F14	Create Premium Duplicate Balanced. Rename Premium Dual Boiler. Also select Milk Temperature Probe and Adaptive Brew Profile; choose 400V. Validate and derive. Why: Third alternative. Done when: Valid and Current.
F15	Audit 100% models For each configuration open 100% Realization, inspect included/excluded/modified items and Applied variation audit, then read Derivation Summary. Why: A green badge alone does not explain the transformation. Done when: Every intended effect is visible.

G. Define KPIs, simulate, compare and decide

G1	Create/select KPIs Open Parameters and KPIs > KPI definitions. Create or select Total Mass, Estimated total cost, Manufacturing Lead Time and Throughput Proxy. Link objectives, units, directions and weights. Why: These are the common alternative measures. Done when: Four valid KPI definitions exist.
G2	Complete variant inputs Add configuration-applicable mass/cost/power inputs with provenance. Keep blank applicability only for values shared by every configuration. Why: The same owner can carry different values per alternative. Done when: Each alternative resolves a numeric value.
G3	Run three simulations For each configuration, choose Configured 100% model, select the same four KPI definitions, name the run and click Validate and run. Why: Trade evidence must be current and comparable. Done when: Three current complete runs exist.
G4	Create Trade Study Open Architecture Trade Study. Create Coffee-machine architecture selection and enter the baseline question. Why: Frames the decision. Done when: Study is active.

G5	Select scope Check intended use cases, needs and objectives. Confirm automatically traced mandatory requirements. Why: Feasibility comes from trace-derived requirements. Done when: No scope warning remains.
G6	Link root and axes Link Coffee Machine Product Line and select/create the loading, brewing and sensing axes. Why: Defines explored design space. Done when: At least one valid axis is selected.
G7	Select KPI set Select the same four objective-linked KPIs and confirm positive total weight. Why: Defines preference calculation. Done when: All four are eligible.
G8	Select alternatives Open Step 2. Check Essential, Balanced and Premium. Use latest current run if shown. Why: Only ready alternatives are enabled. Done when: 3/2 minimum selected.
G9	Run comparison Click Run comparison. Resolve readiness errors at their source. Do not override requirement feasibility. Why: Creates immutable evidence. Done when: A saved result appears.
G10	Inspect manager and expert Read feasibility/ranking in Manager view. Switch to Expert view and inspect raw/normalized values, weights, contributions, formulas, run/derivation IDs and requirement evidence. Why: A score must be traceable. Done when: No unexplained driver remains.
G11	Approve decision Open Step 3, create decision, choose a feasible alternative, enter owner/rationale, confirm baseline effect, confirm selection and approve. Why: The user, not the algorithm, establishes the baseline. Done when: Decision is approved.
G12	Deliver Run validation, create snapshot, save complete JSON, export selected evidence and generate PDF report. Why: Completes audit and recovery path. Done when: Reloadable JSON plus review outputs exist.

Part XI - Complete recipes for each overall scope

Use exactly one of these recipes for a new real project. Each recipe is complete; do not assume that reading another scope supplied a missing step.

Chapter 1 - Architecture building

1	Create and scope project Create a new project, select Architecture building, and save an initial complete JSON. Why: Establishes project identity and route map. Done when: Dashboard shows two problem and two solution steps.
2	Define mission and stakeholders Create mission and stakeholders; link mission with hasStakeholder; designate exactly one system of interest. Why: Required intent and scope foundation. Done when: Problem Step 1 progresses.
3	Define needs and objectives Create at least one of each and link from stakeholders using hasNeed and hasObjective. Why: Requirements must have stakeholder intent. Done when: Problem Step 1 completes.
4	Define use cases Create observable use cases, connect stakeholders with involvedIn, and select eligible system-of-interest use cases as working scope. Why: Limits the investigation. Done when: Working-scope rows are selected without errors.
5	Define requirements Create testable requirements and link Need/Objective -> derives -> Requirement. Add verification methods and executable formulas when numeric. Why: Provides traceable constraints. Done when: Problem Step 2 completes.
6	Build product functions For every selected use case, create needed product functions and hasFunction links. Why: Defines delivered behavior. Done when: Every scoped use case has product behavior.

7	Build product technical architecture Create components/interfaces, hierarchies and realizedBy from every product function. Add requirement satisfiedBy traces. Why: Connects behavior to solution structure. Done when: Solution Step 1 completes.
8	Build industrial functions Create process functions, durations, sequence and product item flows; connect relevant use cases with hasFunction. Why: Defines how product is produced/verified. Done when: Every scoped use case has industrial behavior.
9	Build industrial technical architecture Create industrial components/interfaces; connect process function -> realizedBy -> component. Why: Connects industrial behavior to equipment. Done when: Every process function is realized.
10	Allocate resources Create Resource elements and component -> requiresResource -> resource with positive quantity/unit. Never place requiresResource on a function. Why: Maintains architecture allocation semantics. Done when: Solution Step 2 completes.
11	Validate traceability Use graphs, matrices and Model quality. Close mission-to-requirement-to-function-to-component and process-to-resource traces. Why: Finds gaps beyond simple row counts. Done when: No unexplained model errors.
12	Deliver Create snapshot, save complete JSON, and generate selected export/report. Why: Preserves architecture evidence. Done when: Reloadable file and review package exist.

Architecture-building exit criteria

Area	Minimum accepted evidence
Intent	Mission, mission stakeholder, exactly one system of interest, needs and objectives.
Scope	At least one eligible selected use case and derived requirement.
Product	Every scoped use case has product function(s); every product function has realizing product component.
Industrial system	Every scoped use case has process function(s); every process function has realizing industrial component.
Resources	Resources allocated via requiresResource from realizing industrial components.
Quality	Canonical relationships valid and deterministic validation reviewed.

Chapter 2 - Architecture building + simulation

1	Create intent Create mission, stakeholders, exactly one system of interest, needs and objectives with canonical traces. Why: Same complete intent gate as Architecture building. Done when: Problem Step 1 complete.
2	Create scope and requirements Create involved use cases, select working scope, derive testable requirements and add formula bindings where numeric. Why: Simulation results must be judged against intent. Done when: Problem Step 2 complete.
3	Define evaluation KPIs Create at least one objective-linked KPI. Set formula/algorithm, output unit, direction and non-negative weight. Why: Determines which inputs the architecture needs. Done when: Problem Step 3 complete.
4	Build product architecture Create use-case product functions and realizing product components/interfaces with parameters. Why: Provides product calculation sources. Done when: Solution product step complete.
5	Build industrial architecture Create process functions/durations/sequence, industrial components and realization links; allocate resources through components. Why: Provides time, cost and resource evidence. Done when: Solution industrial step complete.
6	Complete parameters For every required numeric input enter value, unit, origin, provenance and applicability. Use exact semantic keys for standard algorithms. Why: Simulation must not replace missing data with undocumented zeros. Done when: Parameter step complete.
7	Validate formulas Preview calculated parameters and KPIs; confirm requirement statuses and units. Why: Catches missing bindings and incompatible units. Done when: All intended results are numeric or intentionally documented unavailable.

8	Run simulation Choose configured 100% if using a configuration, otherwise Architecture-only 150%; select KPIs/dependencies; name and run. Why: Creates immutable evidence. Done when: Current saved run exists.
9	Review result detail Inspect formula/algorithm, sources, assumptions, missing information, warnings, breakdown and critical chain. Why: A number without provenance is insufficient. Done when: No unexplained warning/missing input.
10	Deliver Validate, snapshot, save complete JSON and generate report including simulation details. Why: Preserves reproducible evidence. Done when: Scope dashboard complete and deliverables saved.

Architecture + simulation exit criteria

Area	Minimum accepted evidence
Architecture	Both product and industrial function-to-technical realization chains complete.
Inputs	Required numeric parameters have value/unit/source and correct active applicability.
KPI	Objective-linked definition with valid formula/algorithm, unit, direction and weight.
Requirements	Numeric rules are bound and status is not pending/error.
Run	Current immutable run from intended model context with selected KPI dependencies.
Review	Sources, assumptions, missing information and warnings accepted/documented.

Chapter 3 - Trade study

1	Complete intent/scope Create mission, stakeholder context, exactly one system of interest, working use cases and derived executable requirements. Why: Feasibility begins in Problem Space. Done when: Problem Steps 1-2 complete.
2	Frame study Create/activate a Trade Study, type name and decision question, select use cases, needs and objectives, and inspect traced mandatory requirements. Why: Defines decision boundary. Done when: Study scope complete.
3	Define root/axes/KPIs Link one root, select at least one valid axis, and select objective-linked KPIs with formulas/algorithms, units, direction and positive total weight. Why: Defines design space and evaluation basis. Done when: Problem Step 3 complete.
4	Build 150% architecture Create complete product and industrial functions/components/interfaces/resources and canonical traces for intended solution range. Why: The family source must precede variation. Done when: Solution Step 1 complete.
5	Define feature model Create one root, mandatory/optional/XOR/OR/enumeration features, groups and constraints. Why: Defines valid option semantics. Done when: Feature-model step complete.
6	Map variability Create enabled existence/property/tag variation points with feature expressions, typed conditions, scopes, safe paths and ordered values. Why: Connects features to architecture effects. Done when: Mapping step complete.
7	Create configurations Create at least two named configurations, select features/values, validate and resolve every error. Why: Defines alternatives. Done when: At least two valid active configurations.

8	Derive 100% Explicitly derive each valid configuration and inspect audit/summary. Why: Creates revision-specific physical architecture instances. Done when: At least two Current derivations.
9	Complete variant inputs Ensure every selected KPI input resolves for every alternative through applicability and variation effects. Why: Comparability requires complete numeric evidence. Done when: No alternative-specific missing input.
10	Simulate consistently Run the same selected KPI definitions for every current derivation, including dependencies. Why: Produces comparable immutable evidence. Done when: At least two current complete runs.
11	Select alternatives In Trade Study Step 2, check each Ready configuration-derived alternative and reference latest current run. Why: Binds exact evidence. Done when: At least two selected alternatives.
12	Run comparison Execute; requirement feasibility first, KPI normalization/weighting second. Repair errors at source. Why: Creates immutable result. Done when: Current saved comparison exists.
13	Review manager/expert Review feasible set, raw values and score; then inspect normalized values, contributions, formulas, run/derivation IDs and requirement evidence. Why: Prevents black-box decision making. Done when: Drivers and limitations understood.
14	Approve decision Create draft, choose feasible alternative, record owner/rationale, confirm baseline and approve. Why: The user establishes baseline explicitly. Done when: Approved decision and baseline architecture.
15	Deliver and reload-test Validate, snapshot, save complete JSON, export selected evidence, generate report, and test Load as new project. Why: Preserves the full audit chain. Done when: Portable project and review artifacts verified.

Trade-study exit criteria

Gate	Required state
Problem Space	Trace-derived scope and mandatory executable requirements.
150% source	Product and industrial functional/technical architecture complete.
Feature model	One root, valid hierarchy/cardinality/constraints and selected axes.
Mapping	Every intended variant effect explicitly mapped and enabled.
Alternatives	At least two valid configurations with current explicit derivations.
Evidence	Same selected KPI set in current complete runs for every alternative.
Comparison	Current immutable result with requirement feasibility and transparent contributions.
Decision	Feasible selection, owner, rationale, baseline confirmation and approved status.

Part XII - Impact matrix, troubleshooting and glossary

Use this final part when an edit appears to have an unexpected consequence, a workflow gate is blocked, or a term in the interface is unfamiliar. Start with the impact matrix to decide whether the field changes calculations, model semantics, governance evidence, or only the display. Then use the recovery table to repair the evidence at its source.

Question	Go to
Will this edit change a result or make evidence stale?	Impact matrix
Why is a step, formula, derivation, run or comparison blocked?	Common failures and exact recovery
What does an interface term mean in this workbench?	Glossary
What is the governing operating rule?	Final operating principle

What changes the storyline and what does not

Field/action family	Actual impact	Does it make evidence stale?	Notes
Name/description of ordinary model element	Name changes display/search; description is documentation.	Name/description can increment model basis because canonical source changed.	Stable formula IDs remain.
Element status	Maturity/validation/governance.	May change validation/maturity, not numeric result.	draft/reviewed/approved.
System-of-interest checkbox	Semantic scope and dashboard/trade eligibility.	Yes, scope evidence changes.	Exactly one required.
Working-use-case checkbox	Project semantic scope.	Yes.	Only eligible use cases can be selected.
Canonical relationship	Traceability, architecture and/or calculation chain.	Yes when calculation/scope basis is affected.	Direction and endpoint types matter.
Element owner/source	Documentation and audit.	Normally no numeric change, but stored model revision may change.	Parameter source is separate.
Tags	Search/filter and optional variation target.	Only variation use changes result; editing source can still change revision.	Not a KPI parameter.
Custom attribute	Stored/exported, required validation and variation target.	Can stale derivation if source changes.	Never offered as KPI input.
Parameter value/unit/applicability	Requirement, KPI and simulation evidence.	Yes.	Direct calculation input.
Parameter origin/source/bounds	Provenance and uncertainty context.	Can alter stored evidence/revision; nominal value is unchanged.	Review impact even when numeric result is same.
Process duration	Lead time, cost, demand, utilization and throughput proxy.	Yes.	minute/hour/day; day = 8 h.
Resource hourly rate / requiresResource quantity	Process cost/demand/utilization.	Yes.	Resource relation must originate at industrial component.
Requirement expression/binding	Pass/fail feasibility.	Yes.	Pending/error is not feasible evidence.
KPI formula/algorithm/unit/direction/weight/objective	Simulation/comparison basis and eligibility.	Yes for calculation-affecting settings.	Dependencies must be selected.
Feature selection/value/constant/variation point	100% architecture and variant parameters.	Yes.	Validate and rederive.
Configuration name	Also names linked architecture.	Can affect stored context; physical selections are separate.	No independent architecture-name input.

Study, decision, display and placeholder fields

Field/action family	Actual impact	Does it make evidence stale?	Notes
Trade Study name/question	Stored framing and output labels.	Current simplified implementation marks these non-calculation-affecting.	Rationale/context only; other scope/KPI edits stale results.
Axis description	Documentation.	No numeric effect by itself.	Axis selection/group link is calculation-affecting.
Simulation run name	Historical label.	No; set before run.	Run inputs determine result.
Decision owner/rationale	Governance evidence.	Does not recalculate comparison.	Required rationale for approval.
Search/filter/layout/audience view	Display only.	No.	VIEW.
Placeholder text	No stored value.	No.	HINT until user types.
Demo Checklist checkbox	Temporary reminder only.	No.	Does not affect dashboard or validation.

Common failures and exact recovery

Symptom	Cause to test first	Recovery path
Dashboard step blocked	Read the step message; prerequisite evidence missing.	Click the activity buttons and repair evidence; no manual override exists.
Use case cannot be selected	No involvedIn link from system of interest.	Create System stakeholder -> involvedIn -> Use case, then select working scope.
Requirement pending	Unbound @symbol, null value, KPI unavailable or unit issue.	Use picker to bind exact source, complete value/unit and Save element.
Requirement error	Syntax, more/less than one comparison, incompatible units or division by zero.	Reduce to safe arithmetic plus exactly one comparator.
Custom column missing from KPI selector	This is intentional.	Create a real numeric Parameter on a model element if it must calculate.
Calculated parameter cannot see source	Source is outside same component/transitive refiner hierarchy.	Correct component refines hierarchy or relocate the parameter deliberately.
KPI formula invalid	Bad exact ID, unknown function, incompatible units or divide by zero.	Reinsert tokens through picker and use only documented grammar.
KPI dependency error	Referenced kpi("id") not selected or cycle exists.	Select dependency or simplify dependency graph.
Standard mass/power/cost missing	Semantic key spelling/applicability/value/unit.	Use exact expected key and current configuration applicability.
Process cost too low/empty	Missing duration, realizedBy, requiresResource, quantity or hourlyRate.	Repair the authoritative chain in that order.
Lead time unexpected	Wrong duration unit, missing/extra precedes or parallel critical path.	Inspect sequence table, validation and critical-chain result.
Item-flow handoff error	Component or quantity unit mismatch between produces and consumes.	Align the produced/consumed product component and compatible unit.
Configuration invalid	XOR/OR cardinality, requires/excludes, missing enumeration value.	Read ordered findings and repair one rule at a time.
Variation point error	No target, bad feature ID/expression, unsafe path or missing value rule.	Use pickers, allowed paths and ordered rules; validate again.
Derivation stale	150% model, feature model, mapping or configuration changed.	Validate and explicitly derive a new 100% realization.
Simulation blocked	Configuration/derivation not ready or KPI inputs missing.	Repair Variability first, then Parameters/KPIs, then rerun.
Simulation stale	Source revision/KPI basis changed.	Create a new run; retain old history.
Alternative disabled	No compatible current run for every selected KPI.	Validate config, derive current, run exact KPI set.
Comparison stale	Scope, axes, KPI definitions/settings or alternative evidence changed.	Refresh derivations/runs and create a new immutable comparison.
Decision approval disabled	No feasible selection, empty rationale, baseline confirmation unchecked or result stale.	Complete all four gates with current result.
Loaded file rejected	Invalid schema/references or corrupted JSON.	Download raw input for diagnosis; invalid load does not modify current project.
Wide tables wrap badly on screen	Viewport is narrower than table's intentional minimum width.	Use horizontal scrolling; sticky selected-element column and responsive controls preserve access.

Glossary

Term	Meaning in this website
150% model	Canonical reusable family model containing common and variable solution content.
100% realization	Immutable derived model for one valid configuration and source revision.
Architecture	Linked identity generated from a configuration; common/specific model context.
Binding	Stable link from a readable formula symbol to a parameter or KPI ID.
Calculated parameter	Component-owned parameter computed from same-component/transitive-refiner parameters.
Configuration	Named set of manual/automatic/effective feature selections, typed values and realization scope.
Current	Evidence still matches its referenced source revision and settings.
Derivation	Recorded transformation result from 150% source plus configuration to 100% realization.
Feasibility	Mandatory requirement evaluation status, applied before KPI preference.
Feature	Selectable or structural product-line choice.
FeatureGroup	Non-selectable organization of features; can be synchronized with a trade-study axis.
KPI	Reusable definition of a numeric measure, not the run result itself.
MBPLE	Model-based product-line engineering: feature/configuration/mapping/derivation of variants.
MBSE	Model-based systems engineering: connected intent, requirements, behavior, structure and verification.
Parameter	Typed model-owned engineering value with unit, origin, source and applicability.
Physical architecture	Product components and industrial-system components that realize functions.
Provenance	Recorded source of an input or evidence item.
Semantic key	Stable engineering classification used by standard algorithms, for example mass.
SimulationRun	Immutable selected-model input snapshot plus calculated KPI/algorithm results.
Stale	Retained evidence whose source model/settings no longer match current project state.
System of interest	Exactly one stakeholder designated as the system being designed/assessed.
Trade Study	Traceable decision workflow using feasibility, KPI comparison and explicit approval.
Variation point	Rule mapping feature conditions to existence or property effects on canonical content.
Working scope	Selected eligible use cases that bound the active investigation.

Final operating principle

Model first, calculate second, decide explicitly

A trustworthy result is navigable from mission and stakeholder intent to requirement, function, technical component, parameter/source, variant transformation, saved simulation, feasibility evidence, weighted comparison and approved rationale.

Public page: <https://mbse-mbple-workbench.onrender.com/>

Part XIII - Architect and Modeler answer handbook

This branch-accurate handbook supplements the established guide. It explains what an unfamiliar user should enter, what the answer creates or changes, and when a question is review-only. Dynamic questions repeat for each item in the project.

Handbook index

How to answer Architect questions

- Study setup and intent (AV-A01 to AV-B07)
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Using Modeler view safely

Quick answer-quality checklist

How to answer Architect questions

The number of visible questions changes with the chosen scope and with the items you create. Questions marked "per item" repeat for each stakeholder, use case, requirement, function, component, KPI, feature axis or alternative.

Study setup and intent (AV-A01 to AV-B07)

Question	What to enter or select	What it changes
AV-A01 - project aim	Describe the intended project outcome and why the work is being undertaken.	Stores the authoritative project description before scope selection.
AV-A02 - study scope	Choose Architecture definition, Architecture and simulation, or Trade-off according to the decision you need to support.	Sets how far the guided workflow continues. Existing project data is preserved when scope changes.
AV-A03 - study name	A short unique project name, such as Office coffee-machine concept.	Renames the active project.
AV-A04 - mission	One clear sentence describing the system's purpose and the problem it solves. Avoid listing a preferred solution.	Creates or updates the Mission element.
AV-B01 - stakeholders	Semicolon-separated people or organisations whose concerns matter. Do not enter the system being designed as a stakeholder.	Creates Stakeholder elements and canonical mission-participation links.
AV-A05 - system of interest	Enter the name of the system being designed or analysed.	Creates or updates the dedicated System element and its canonical hasSOI link from the Mission; use cases refer to the System as their subject.
AV-B06/B07 - external systems and roles	Enter external-system names, then describe each system's role or boundary interaction.	Creates External System elements and canonical Mission participatesInMission links; those systems can later receive involvedIn links to use cases.
AV-B02 - stakeholder role	Describe why each stakeholder matters.	Updates documentation only; it creates no relationship and changes no calculation.
AV-B03 - stakeholder needs	Desired outcomes or problems from that stakeholder's viewpoint, separated by semicolons.	Creates Need elements linked to the stakeholder.
AV-B04 - stakeholder objectives	Measurable improvements or targets, separated by semicolons.	Creates Objective elements linked to the stakeholder.
AV-B05 - intent recap	Confirm only after mission, stakeholders, needs and objectives describe the study correctly.	Records review; it does not override validation findings.

Use cases and working scope (AV-C01 to AV-C05)

Question	What to enter or select	What it changes
AV-C01 - use cases	Short action-oriented scenarios, separated by semicolons, such as Prepare beverage; Perform maintenance.	Creates Use Case elements.
AV-C02 - participants	For each use case, select all involved stakeholders, including the system of interest where applicable.	Creates use-case participation links.
AV-C03 - addressed intent	Select only the needs/objectives actually addressed by that use case.	Creates canonical addresses relationships.
AV-C04 - analysis scope	Select the use cases this study will analyse. Only eligible cases involving the system of interest are offered.	Sets the persistent working use-case scope.
AV-C05 - scope recap	Check participants, addressed intent and active cases.	Records review and exposes missing scope evidence.

Requirements (AV-D01 to AV-D10)

Question	What to enter or select	What it changes
AV-D01/D02 - requirements from needs/objectives	Testable statements. Reuse an existing requirement when it expresses the same obligation; otherwise add a new one.	Creates System Requirement elements and canonical derives links.
AV-D03 - relevant use cases	Select the active use cases in which each requirement applies.	Stores guided applicability context.
AV-D04 - quantitative or qualitative	Select quantitative when a number and comparison can test the requirement; qualitative when inspection, demonstration or another non-numeric judgement is appropriate.	Chooses the formula or verification path.
AV-D05 - quantitative check	Enter measured property, semantic key, operator, target and unit. Example: mass, semantic key mass, <=, 12, kg.	Creates a formal requirement formula intent. It does not invent an actual value.
AV-D06 - actual-value source	Select the function/component that owns the measured property, or leave ownership for later architecture work.	Binds the formula intent to an authoritative model owner when one exists.
AV-D07 - preliminary value	Enter a value only when supported; include unit, source, origin and optional uncertainty/range. Otherwise leave it pending.	Creates or updates the owner's parameter and evidence fields.
AV-D08 - qualitative verification	Name an analysis, inspection, demonstration or test and describe what will be checked.	Creates or updates a Verification Method.
AV-D09 - importance	Mandatory means failure makes an alternative infeasible; Important and Desirable express lower decision priority.	Sets the requirement class.
AV-D10 - requirement recap	Confirm origins, wording, formula/verification path and ownership.	Records review without forcing a passing result.

Trade framing and early KPI intent (Trade-off only: AV-T01 to AV-T06, then AV-I01 to AV-I07)

Question group	What to enter or select	What it changes
AV-T01/T02 - study and decision	A short Trade Study name and one explicit decision question comparing feasible concepts.	Creates/updates the active Trade Study and its open decision context.
AV-T03 - common content	Capabilities or solution elements required in every alternative; exclude things that vary.	Proposes mandatory common features.
AV-T04 - variation areas	Independent choice areas such as brewing technology or interface type, not whole alternatives.	Proposes reusable variability axes.
AV-T05 - concepts	At least two alternative names with short descriptions.	Creates configuration intents used later.
AV-T06 - framing recap	Verify decision, common content, variation axes and concepts.	Records review.
AV-I01 - evaluation results	Select only KPIs needed to judge the objectives.	Creates/selects canonical KPI definitions; it does not run analysis.
AV-I02 - KPI purpose	Link each KPI to the needs/objectives it measures.	Updates KPI objective links.
AV-I03 - calculation method	Use a standard algorithm when its meaning and inputs fit; otherwise choose a guided formula.	Sets the KPI calculation mode.
AV-I06 - interpretation	Enter output unit, minimize/maximize direction, target and optional acceptable bounds.	Defines how the KPI is read.
AV-I07 - global weight	Enter a finite non-negative weight. At least one selected KPI must be greater than zero.	Sets the reusable KPI default weight.

Product behavior and product architecture (AV-E01 to AV-F06)

Question group	What to enter or select	What it changes
AV-E01/E02 - product functions and order	For each use case, enter product functions in normal execution order; then correct the proposed sequence if needed.	Creates Product Function elements, use-case hasFunction links, a contextual sequence and cycle-free precedes links.
AV-E03 - function satisfaction	Select requirements directly helped by each function; leaving a function unlinked is allowed when justified.	Creates requirement satisfiedBy links.
AV-E04 - product-behavior recap	Check use-case coverage, sequence and requirement links.	Records review.
AV-F01 - realizing components	Name one or more physical/logical product components that perform each function.	Creates Product Component elements and realizedBy links.
AV-F02 - component satisfaction	Select requirements directly satisfied by each component.	Creates requirement satisfiedBy links.
AV-F03/F04 - parameter owner/value	Select the authoritative product function/component for a pending measured property; then enter value, unit, source and origin if known.	Creates or reuses one canonical parameter and binds the requirement formula.
AV-F05 - product interaction	When useful, select another product component, name the interface and describe exchanged material, energy or information.	Creates Interface content and canonical links.
AV-F06 - product-architecture recap	Check function realization, interfaces, parameters and satisfaction.	Records review and exposes gaps.

Industrial behavior and industrial architecture (AV-G01 to AV-G12)

Question group	What to enter or select	What it changes
AV-G01/G02 - process functions and order	For each use case, enter industrial-system process functions such as positioning, fastening or inspection, then verify their order. Leave blank for a genuinely product-only case.	Creates Process Function elements, hasFunction, sequences and precedes links.
AV-G03 - process satisfaction	Select requirements each process function helps satisfy.	Creates requirement satisfiedBy links.
AV-G04 - realizing equipment	Name the workstation, equipment or plant element that performs each process function.	Creates Industrial System Components and realizedBy links.
AV-G05/G05A/G05B - industrial evidence	Select satisfied requirements; resolve any property owned by an industrial function/component and enter evidence if known.	Creates satisfaction links and canonical parameters.
AV-G06 - duration	Enter the authoritative process duration, time unit, source, origin and optional uncertainty.	Updates process metadata used by lead-time, cost and resource calculations.
AV-G07/G08 - consumed/produced product	Select product components and enter positive quantity and unit.	Creates canonical consumes and produces flows.
AV-G09/G10 - resources	Name required people, roles, skills, tools, machines, software or facilities; then enter quantity, unit, optional rate, capacity and availability.	Creates Resource elements and industrial-component requiresResource links with quantitative metadata.
AV-G11 - industrial interaction	Add only meaningful material, energy or information exchanges between industrial components.	Creates interface content and links.
AV-G12 - industrial recap	Check realization, sequence, handoffs, duration and resources.	Records review and exposes model-quality gaps.

Traceability and readiness (AV-H01 to AV-H04)

Question	What to enter or select	What it changes
AV-H01 - verification	For every requirement, identify a credible analysis, inspection, demonstration or test; optionally allocate a process function.	Creates/updates Verification Methods and verifies/allocation links.
AV-H02 - satisfying evidence	Select at least one compatible function, technical component or existing owned parameter. Do not create a second parameter here.	Creates satisfaction links or binds the selected parameter and its owner to the requirement formula.
AV-H03 - evidence credibility	Confirm only when value, target, unit, source, origin and verification are credible; otherwise mark pending.	Records an explicit evidence-review state.
AV-H04 - architecture readiness	Review intent, behavior, realization, satisfaction, flows, resources and verification.	Calculates readiness from the model; confirmation cannot hide blockers.

Simulation questions (Architecture and simulation: AV-I01 to AV-I12)

After choosing KPIs and their purpose/calculation/interpretation as described above:

Question group	What to enter or select	What it changes
AV-I04/I05 - exact calculation	Confirm the inputs used by a standard algorithm, or build a formula by inserting named parameters/KPIs.	Stores no duplicate values; formulas keep immutable IDs.
AV-I08 - missing inputs	Select every authoritative contributing owner and enter separate value, unit, source, origin and optional uncertainty.	Creates/updates canonical parameters used by the KPI.
AV-I11 - run	Enter a meaningful run name and execute.	Validates inputs and creates one immutable Simulation Run.
AV-I12 - Simulation Results	Confirm only after KPI results, requirement status, warnings and sources match the intended analysis.	Records review of current evidence. The current architecture is selected automatically.

Variability and 100% configurations (Trade-off only: AV-J01 to AV-K05)

Question group	What to enter or select	What it changes
AV-J01/J02 - family and common features	Name the complete product line and confirm mandatory capabilities shared by all alternatives.	Creates one Root Feature and mandatory child features.
AV-J03/J04 - axes and choices	For each variation area, list choices and select XOR, OR, optional or typed mode; list allowed values for typed axes.	Creates synchronized Variability Axes, FeatureGroups and Features.
AV-J05 - constraints	Add only necessary requires or excludes rules.	Prevents invalid feature selections.
AV-J06 - element applicability	Map only genuinely variable requirements/functions/components/processes/resources/verification methods to a feature; unlisted elements remain common.	Creates existence applicability using the existing variation mechanism.
AV-J07/J08 - property variation and domain	Map a feature to a changed existing parameter value and select affected model domains when needed.	Creates typed Variation Points with exact target and realization scope.
AV-J09 - 150% recap	Check family completeness, hierarchy, constraints, applicability and property changes.	Records review.
AV-K01 - configuration	For each concept, choose one value for every required group and any optional features.	Creates/updates a Configuration.
AV-K02/K03 - invalid selection	Apply a proposed required-feature correction or return to the choices.	Keeps invalid drafts visible but blocks derivation/simulation until valid.
AV-K04 - derive	Review included, excluded and modified content, then derive.	Creates immutable 100% realization evidence without changing the 150% source.
AV-K05 - realization recap	Compare selected features with included/excluded elements and applied property changes.	Confirms the current derivation.

Simulate, compare and decide (Trade-off only: AV-S01 to AV-L10)

Question group	What to enter or select	What it changes
AV-S01 - readiness	Verify every intended alternative has a valid configuration and current 100% derivation.	Identifies blockers; it creates no run.
AV-S03/S04 - comparable simulation	Run the same KPI set for every alternative, then review coverage, units, warnings and sources side by side.	Atomically creates one immutable configured Simulation Run per alternative; no partial batch is stored.
AV-L01/L02/L03 - comparison set	Select at least two eligible alternatives, mandatory feasibility requirements and comparable KPIs.	Sets the active study's evidence basis.
AV-L04/L05 - priorities and limits	Set study-specific KPI weights, preferred direction and optional warning/hard thresholds.	Updates study settings without changing global KPI definitions.
AV-L06/L07/L08 - compare	Resolve blocking comparability gaps, run the calculation and review feasibility, weighted scores, ties and sensitivity.	Stores an immutable Comparison Result. Infeasible alternatives remain visible but cannot be preferred.
AV-L09/L10 - decision	Explicitly select a feasible baseline, record rationale, owner, status/date and confirm approval.	Creates/updates the Decision and changes the baseline only after explicit approval.

Final overview (AV-M02 to AV-M06)

The final pages are reviews, not new data-entry steps. They summarise the study question, model content, stakeholder traceability, requirements, architecture, current simulation and - for Trade-off - alternatives and decision. The completion banner is green only when required answers and evidence are complete and reviewed; amber means pending evidence, warnings, skipped/incomplete answers or answers needing review; red means a failed requirement or another blocking model finding.

Using Modeler view safely

Use Project Workflow to find the next activity. Each Model section edits canonical elements and relationships through Table, Graph, Sequence, Matrix, Section Recap, Model Digital Thread and Model Quality views as applicable. Sequence overview displays the actual precedes network, including parallel branches and joins. The table keeps core fields visible and reveals status, tags, custom attributes and connections through **More columns**. Element editing always opens in the right-side drawer. Interfaces and Traceability are separate sections. Variability edits the feature model, variation points, configurations and 150%/100% realizations; organizational feature groups appear in the same hierarchy table as features. Parameters and KPIs edits authoritative engineering inputs and formulas. Simulation creates immutable runs. Architecture Trade Study frames, compares and decides. Project Recap follows the complete model from mission through architecture, evidence and decision. Delivery saves projects, snapshots and exports. Model Quality collects project-wide findings and links back to affected content.

Section recaps do not prescribe arbitrary element counts. They derive their expectations from current model rules and upstream content-for example, every defined function needs a realizing component and every selected use case needs behavior. Guided examples may state expected counts as walkthrough aids, but those counts are not project validation rules.

When an Architect question is unclear, switch to Modeler view to inspect the relevant table, matrix or graph, then return. If a Modeler edit changes content used by a confirmed answer, the affected question and recap are marked for review; unrelated documentation edits stay confirmed. Never duplicate an existing parameter merely to satisfy a question: identify its authoritative owner and reuse the canonical value.

Quick answer-quality checklist

- Use stakeholder language for needs and measurable language for objectives.
- Write requirements so a verification method or formula can decide them.
- Give every numeric value a unit, source and value origin.
- Keep product functions distinct from the product components that realize them.
- Keep industrial process functions distinct from workstations/equipment that realize them.
- Allocate resources to the industrial component that requires them.
- Treat optional content and changed values as variability; keep common content unconditioned.
- Compare only current 100% derivations with current, complete and comparable simulation evidence.
- Record a decision rationale; do not treat the highest score as automatic approval.